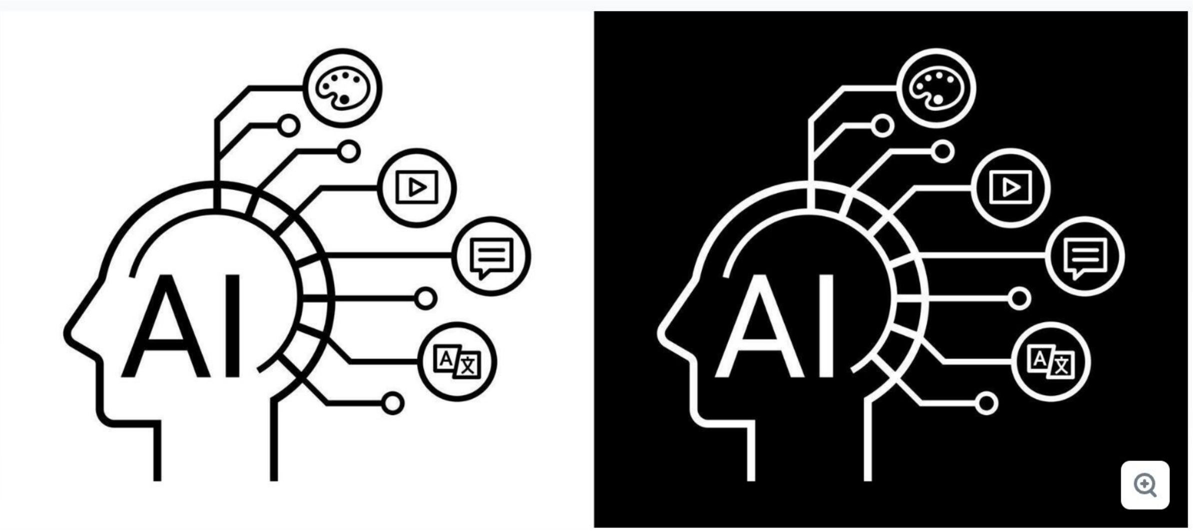
AIMS 5740
Generative Artificial Intelligence

(2026 Term 2)



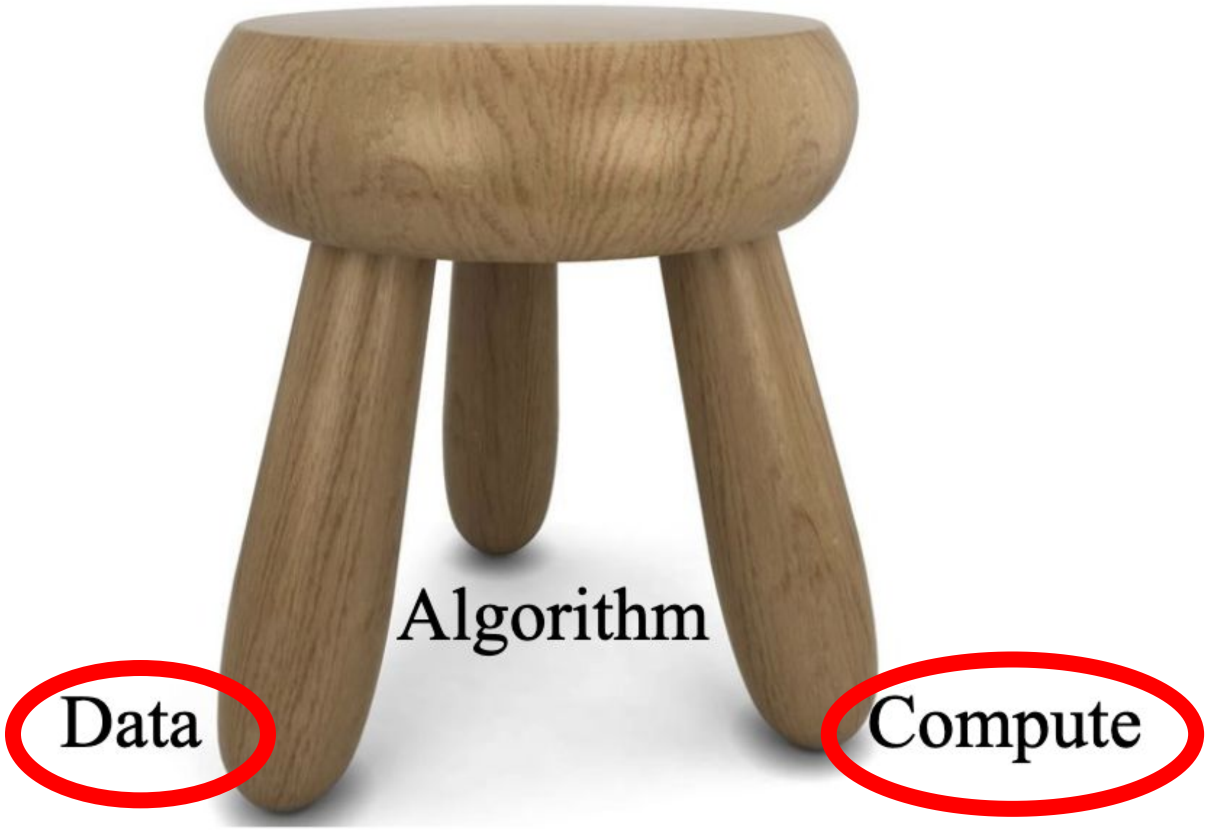
Computer Science & Engineering
The Chinese University of Hong Kong

Announcement

- You can submit your first assignment by Mar. 8.
- Register the form if you can not find team members (before Feb. 28).
- The mid-term quiz is scheduled on Mar. 18 (you can only bring the course slides)

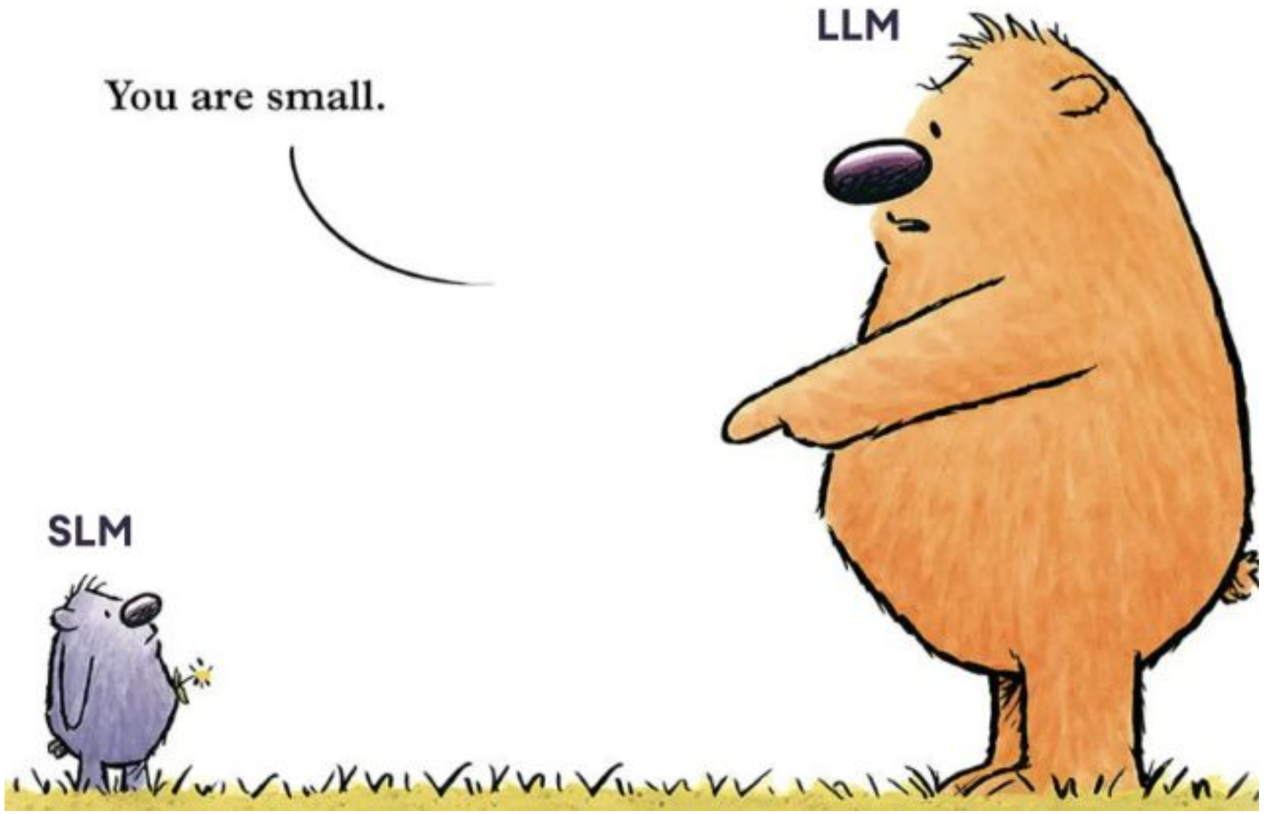
Generative AI

- A comprehensive view of both data and compute.



Model Scaling

- Can we use the results from small models to **predict** the results of large models?



How Large are LLMs?

- How to choose this combination of training tokens / model parameters?

Comparison of some recent **large language models** (LLMs)

Model	Creators	Year of release	Training Data (# tokens)	Model Size (# parameters)
GPT-2	OpenAI	2019	~10 billion (40Gb)	1.5 billion
GPT-3 (cf. ChatGPT)	OpenAI	2020	300 billion	175 billion
PaLM	Google	2022	780 billion	540 billion
Chinchilla	DeepMind	2022	1.4 trillion	70 billion
LaMDA (cf. Bard)	Google	2022	1.56 trillion	137 billion
LLaMA	Meta	2023	1.4 trillion	65 billion
LLaMA-2	Meta	2023	2 trillion	70 billion
GPT-4	OpenAI	2023	?	? (1.76 trillion)
Gemini (Ultra)	Google	2023	?	? (1.5 trillion)
LLaMA-3	Meta	2024	15 trillion	405 billion

How much did it cost to train LLaMa?

Llama-1

When training a 65B-parameter model, our code processes around 380 tokens/sec/GPU on 2048 A100 GPU with 80GB of RAM. This means that training over our dataset containing 1.4T tokens takes approximately 21 days.

Llama-2

		Time (GPU hours)	Power Consumption (W)	Carbon Emitted (tCO_2eq)
LLAMA 2	7B	184320	400	31.22
	13B	368640	400	62.44
	34B	1038336	350	153.90
	70B	1720320	400	291.42
Total		3311616		539.00

GPU Costs

- modern GPUs cost around $15k
- the cost of a cloud GPU per hour ranges $1-$4
- 700W = 0.7 kWh ➔ $0.084 per hour

How much did it cost to train LLaMa?

Llama-3

Compute. Llama 3 405B is trained on up to 16K H100 GPUs, each running at 700W TDP with 80GB HBM3, using Meta's Grand Teton AI server platform (Matt Bowman, 2022). Each server is equipped with eight GPUs and two CPUs. Within a server, the eight GPUs are connected via NVLink. Training jobs are scheduled

	Time (GPU hours)	Power Consumption (W)	Carbon Emitted(tCO2eq)
Llama 3 8B	1.3M	700	390
Llama 3 70B	6.4M	700	1900

GPU Costs

- modern GPUs cost around $15k
- the cost of a cloud GPU per hour ranges $1-$4
- 700W = 0.7 kWh ➔ $0.084 per hour

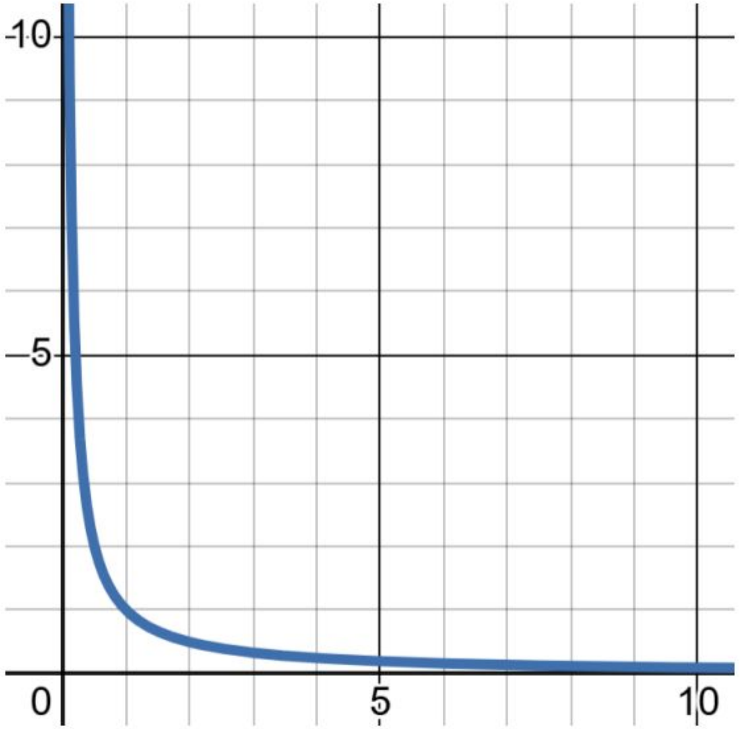
Question: How much did Llama-3 70B cost to train?

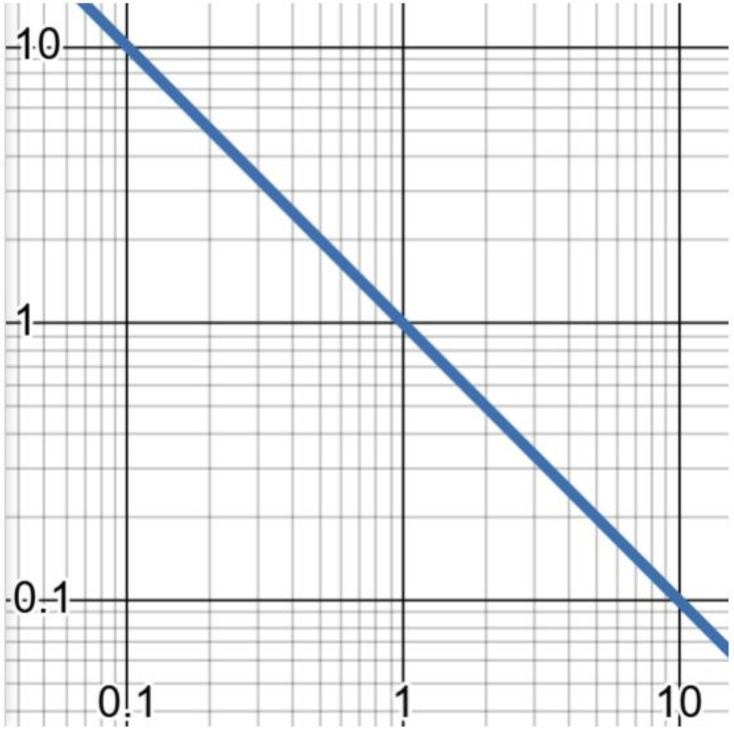
Answer:

Pow Law

- Most scaling laws for LLMs assume we are fitting a power law function
- Definition: a power law function has the form

$$f(x) = cx^{-k}$$





Pow Law

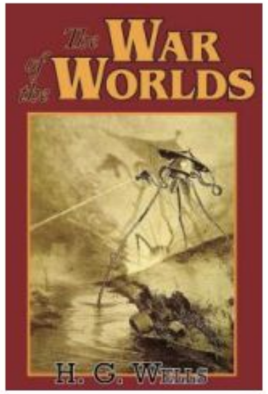
- Example:

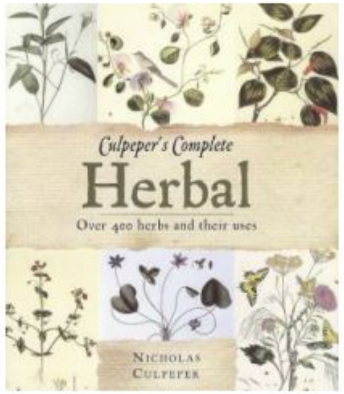
– Zipf's law states the the **n-th most common word** in a corpus appears twice as frequently as the (n+1)-st most common word

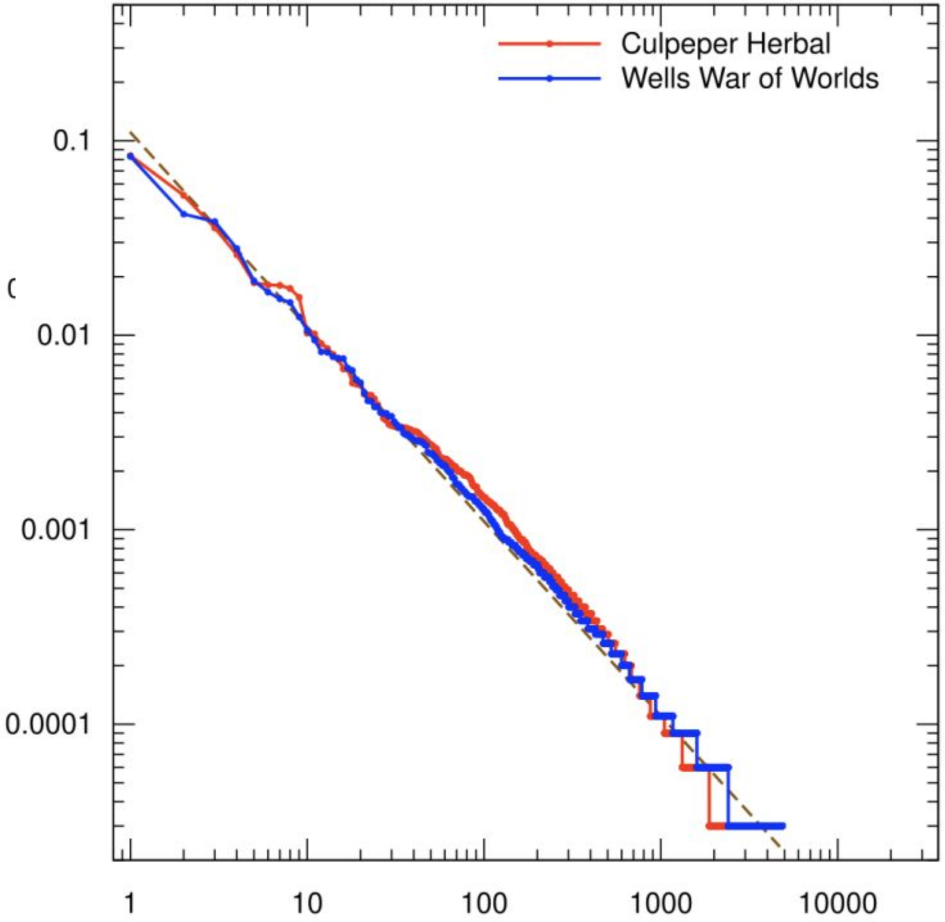
– The Zipf-Mandelbrot law:

$$\text{frequency} \propto \frac{1}{(\text{rank} + b)^a}$$

where a, b are fitted parameters, with $a \approx 1$, and $b \approx 2.7$.[1]



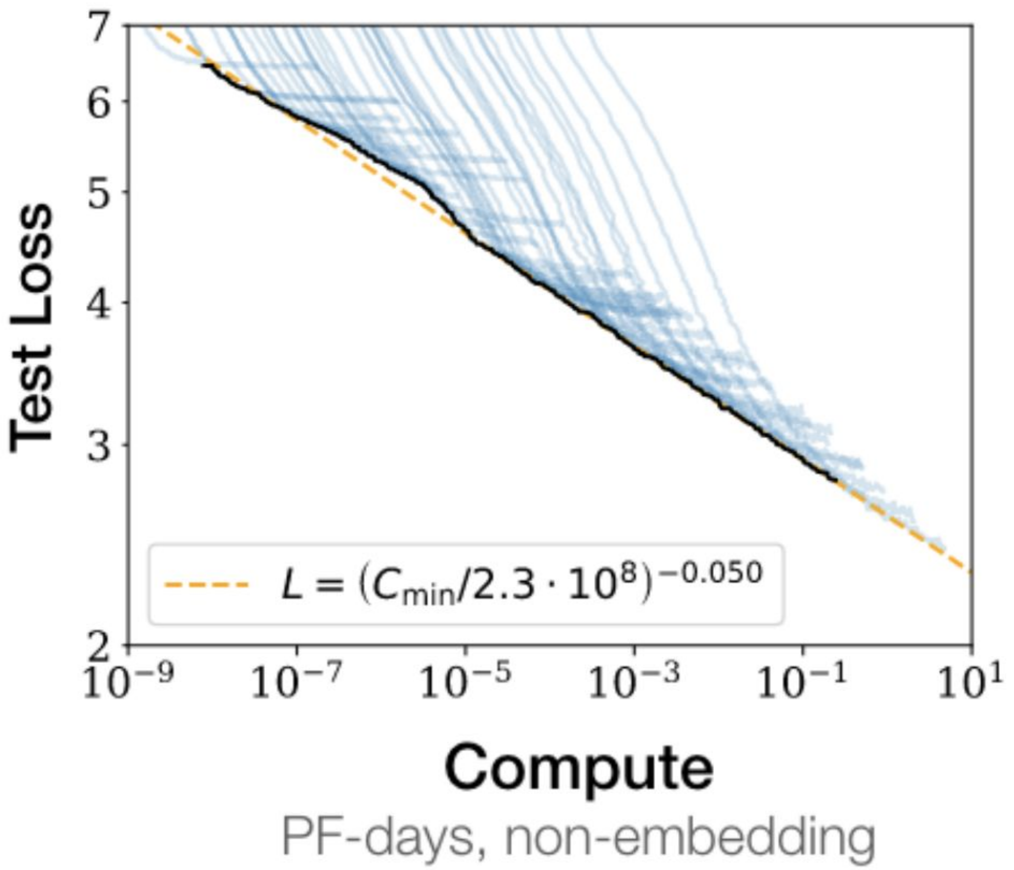




Scaling Laws: Kaplan et al. (2020)

Experimental Design

- Varied a number LLM hyperparams
  - N: # parameters (768 – 1.5B)
  - D: # tokens (22M – 23B)
  - C: # FLOPS
  - model (depth, width, # heads, d_{model})
  - context length (1024 or less)
  - batch size (2^{19} or less)
- Evaluated test loss of each model



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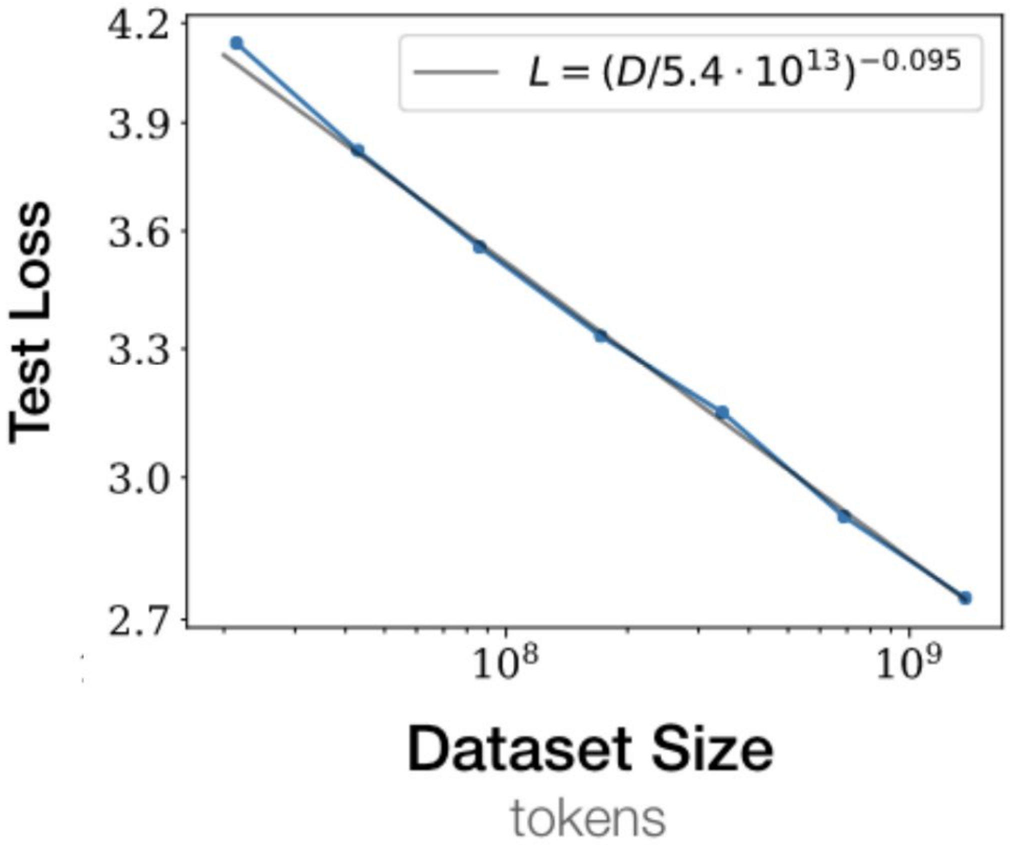

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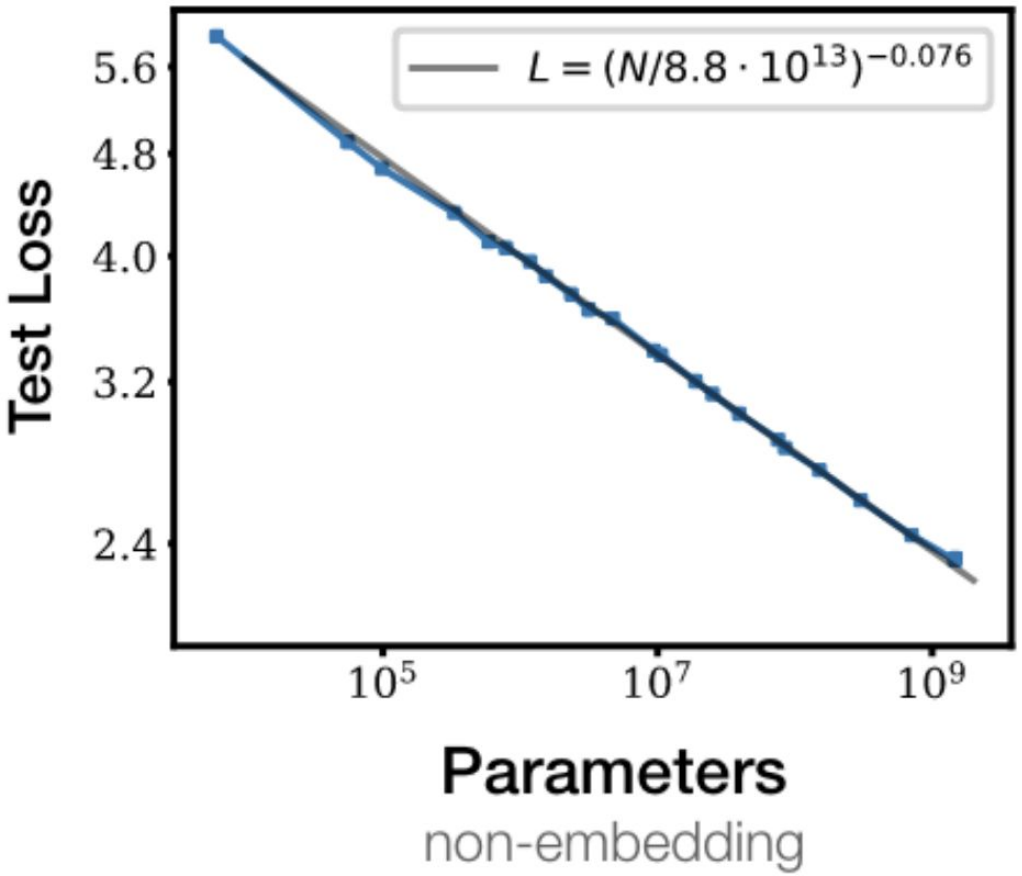
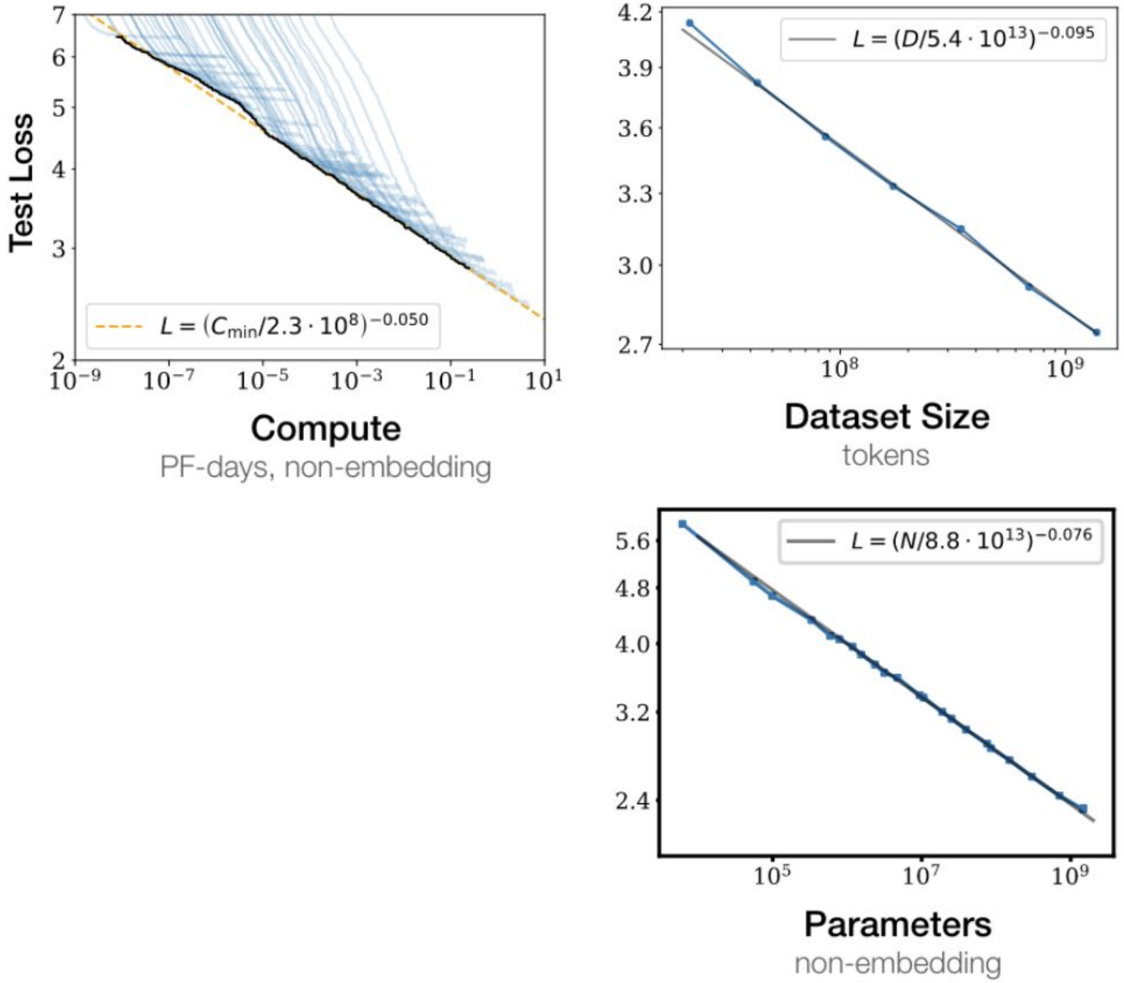


Scaling Laws: Kaplan et al. (2020)

Key takeaways:

1. **three quantities dominate: N = # parameters, D = # tokens, C = # FLOPS**
2. model shape doesn't matter very much
3. performance improves as long as we increase both N and D
4. training loss curves follow predictable power laws
5. larger models are more sample efficient
6. convergence is not critical for good performance
7. best batch size follows a power law (and is huge: 1-2M tokens)



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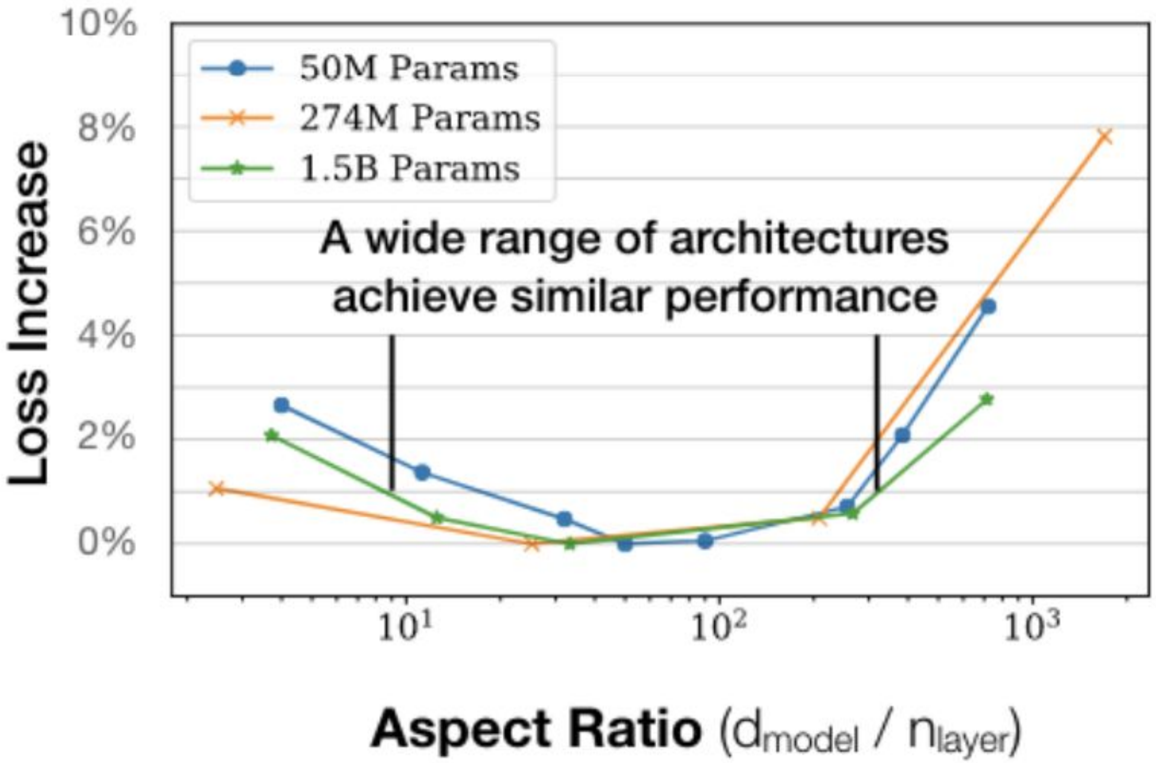

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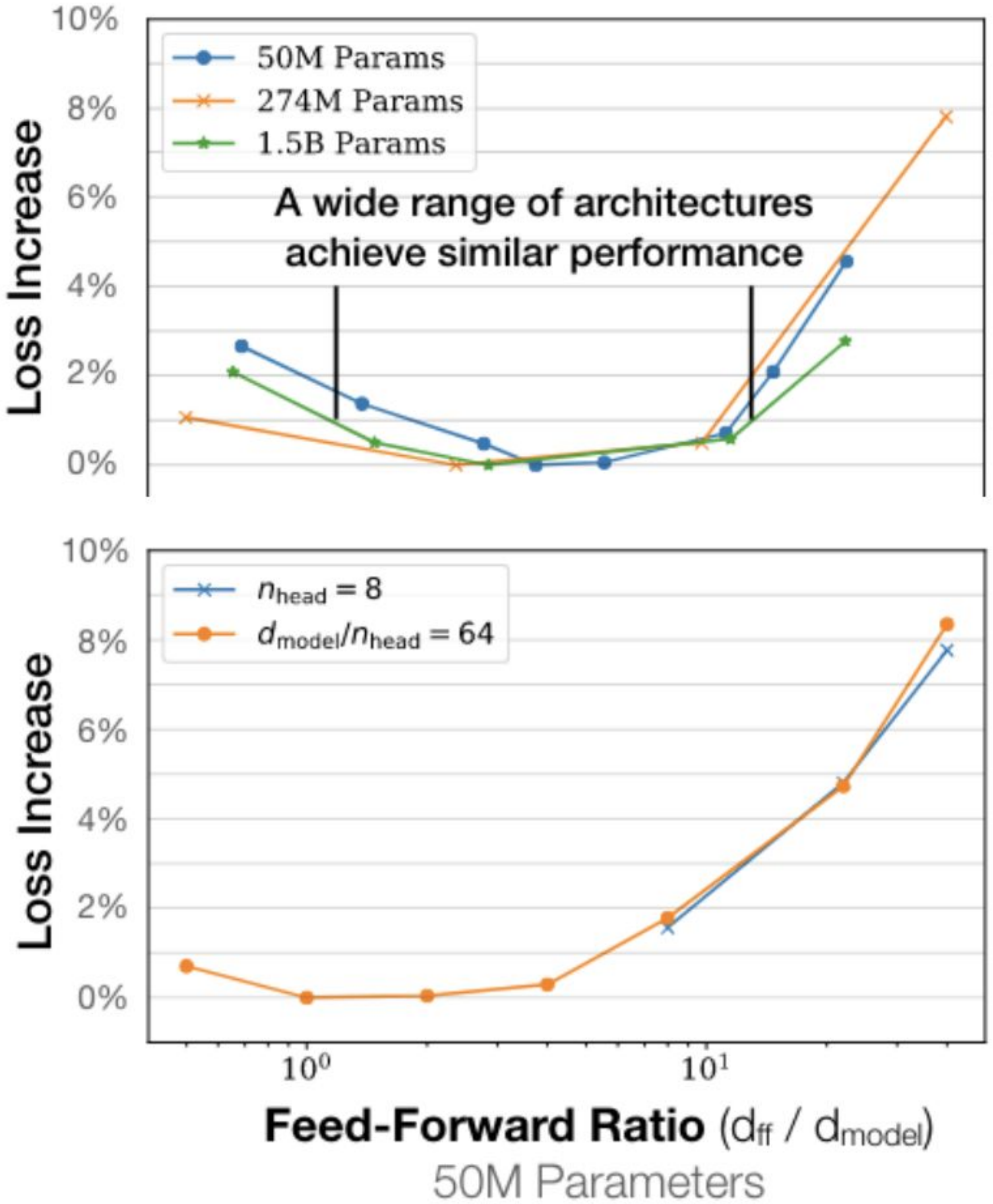

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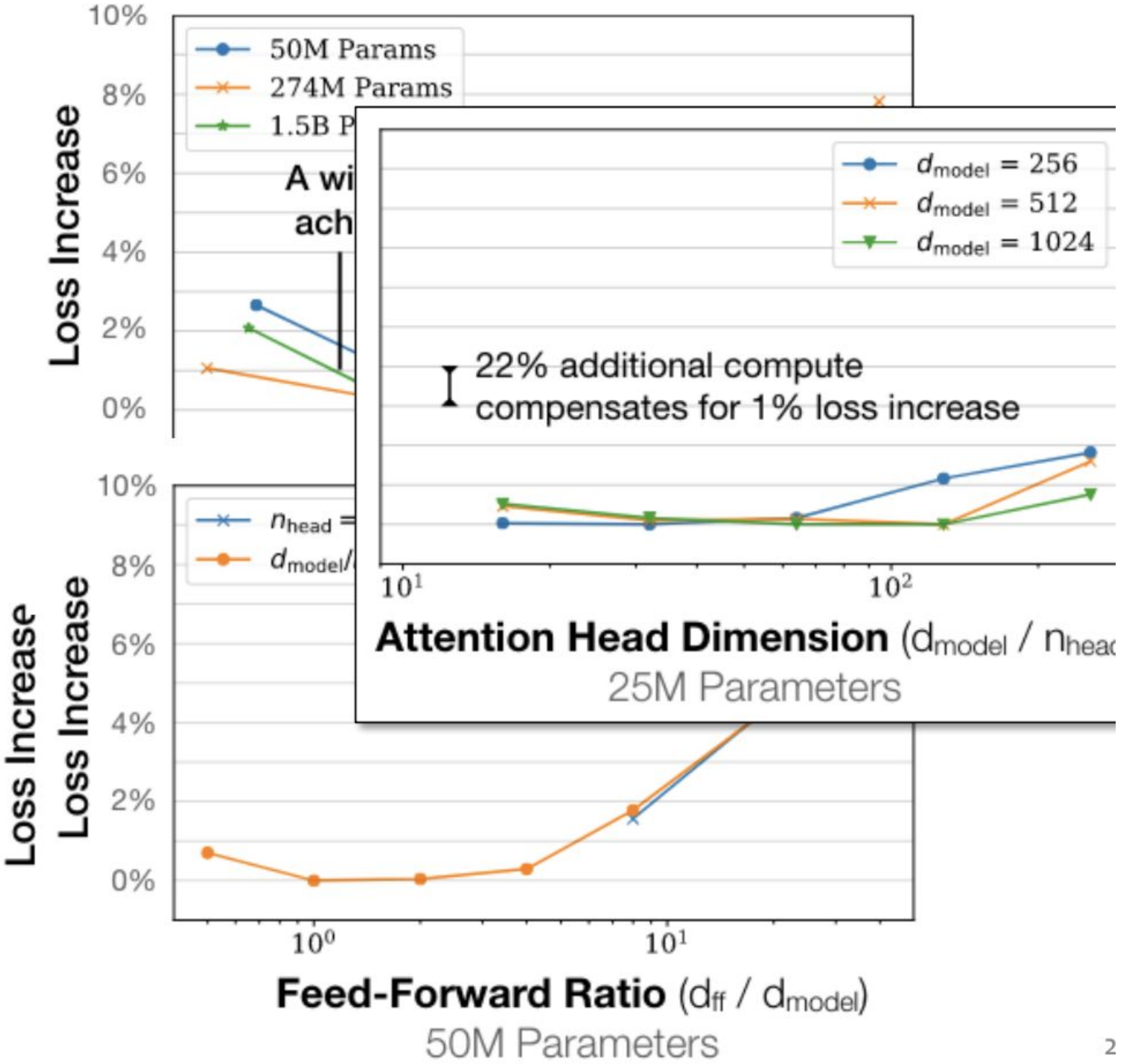

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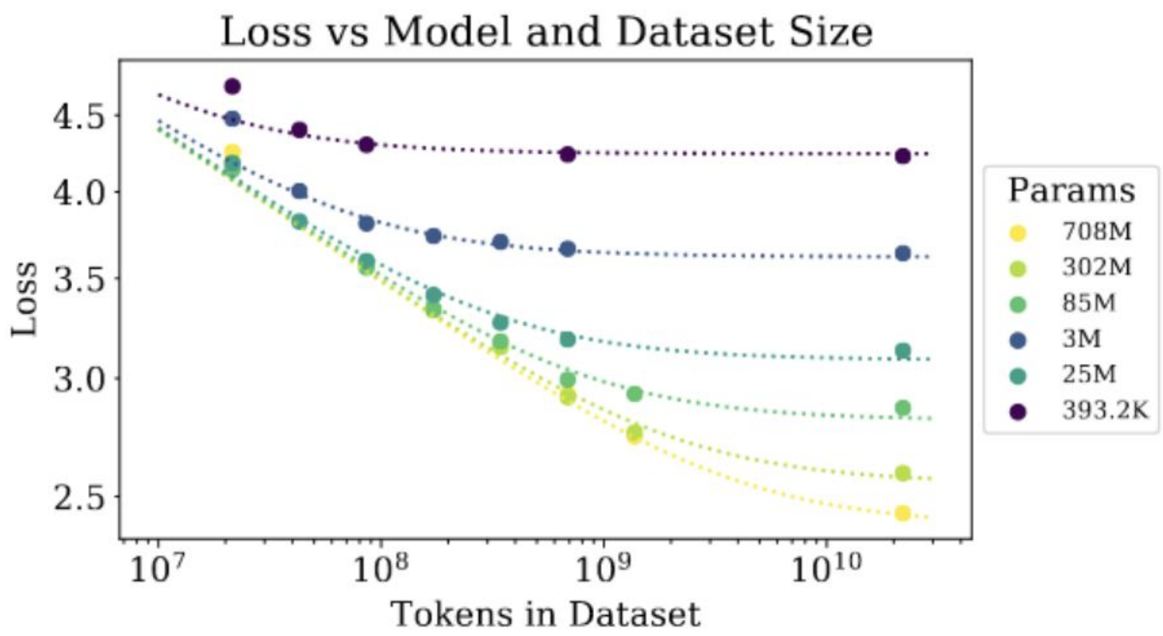

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5. **larger models are more sample efficient**
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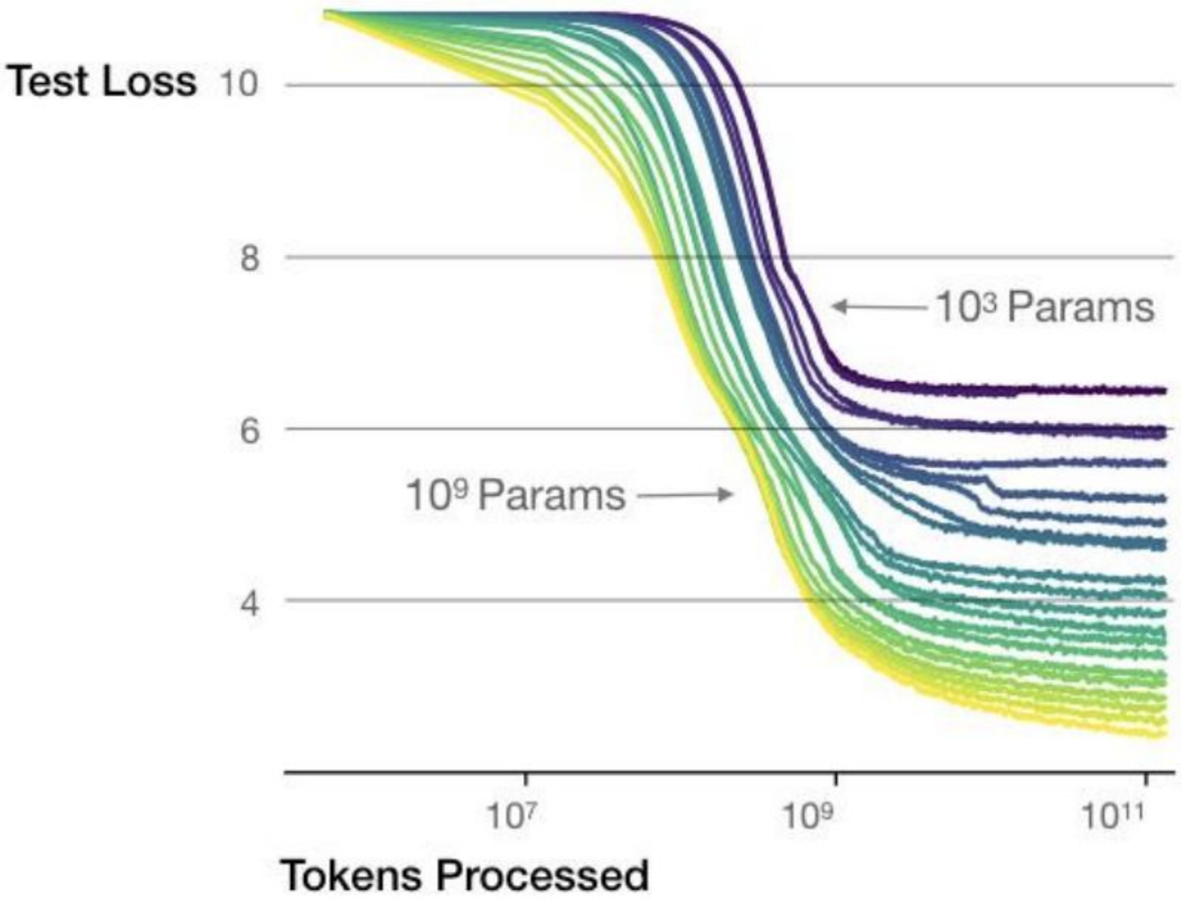
Larger models require **fewer samples** to reach the same performance



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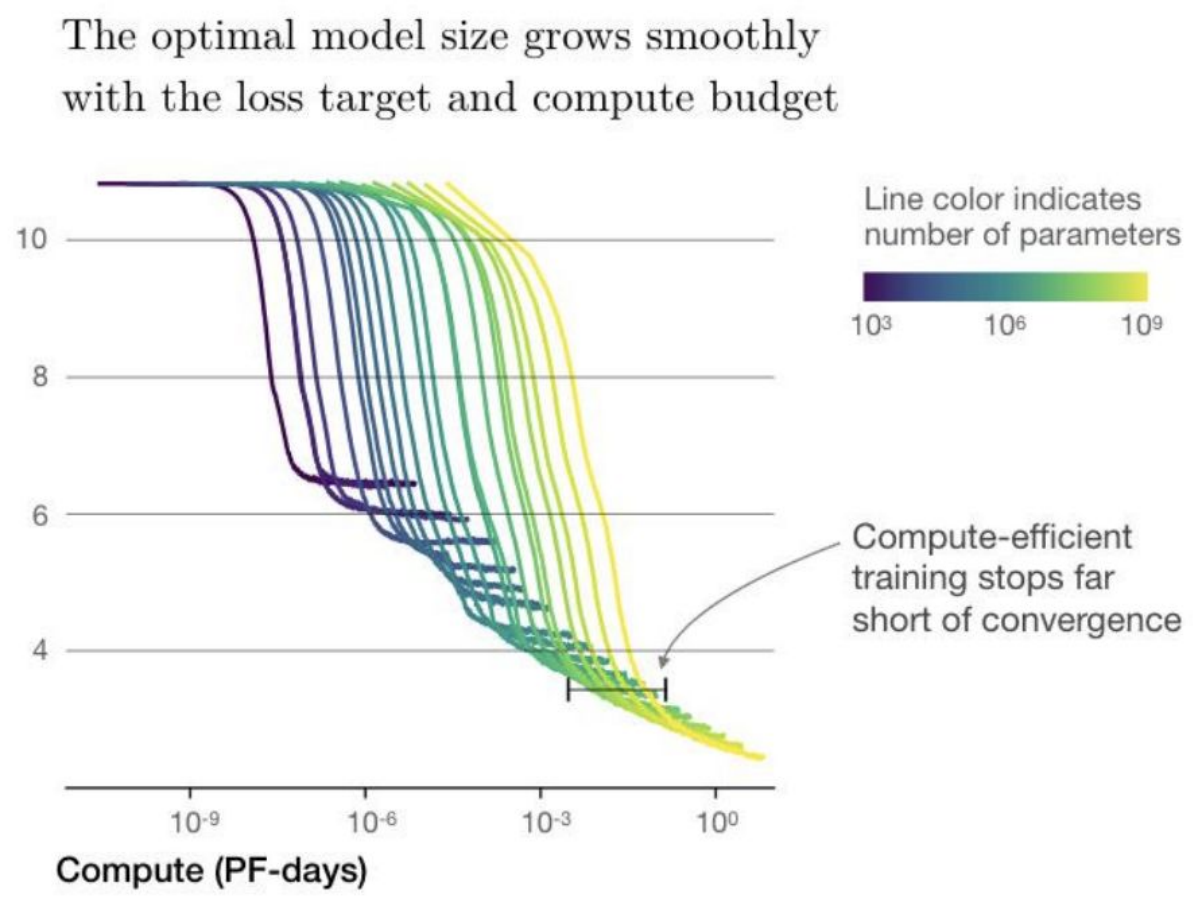

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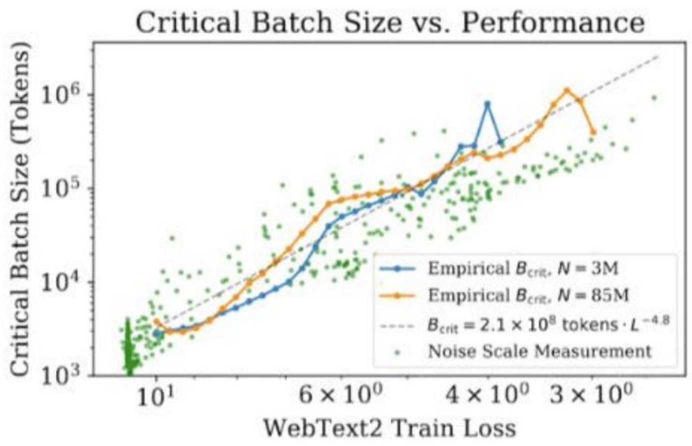

Figure 10 The critical batch size B_{crit} follows a power law in the loss as performance increase, and does not depend directly on the model size. We find that the critical batch size approximately doubles for every 13% decrease in loss. B_{crit} is measured empirically from the data shown in Figure 18, but it is also roughly predicted by the gradient noise scale, as in [MKAT18].

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"every time we increase the model size 8x, we only need to increase the data by roughly 5x to avoid a penalty."

$$L(N, D) = \left[\left(\frac{N_c}{N}\right)^{\frac{\alpha_N}{\alpha_D}} + \frac{D_c}{D}\right]^{\alpha_D}$$

Parameter	α_N	α_D	N_c	D_c
Value	0.076	0.103	6.4×10^{13}	1.8×10^{13}

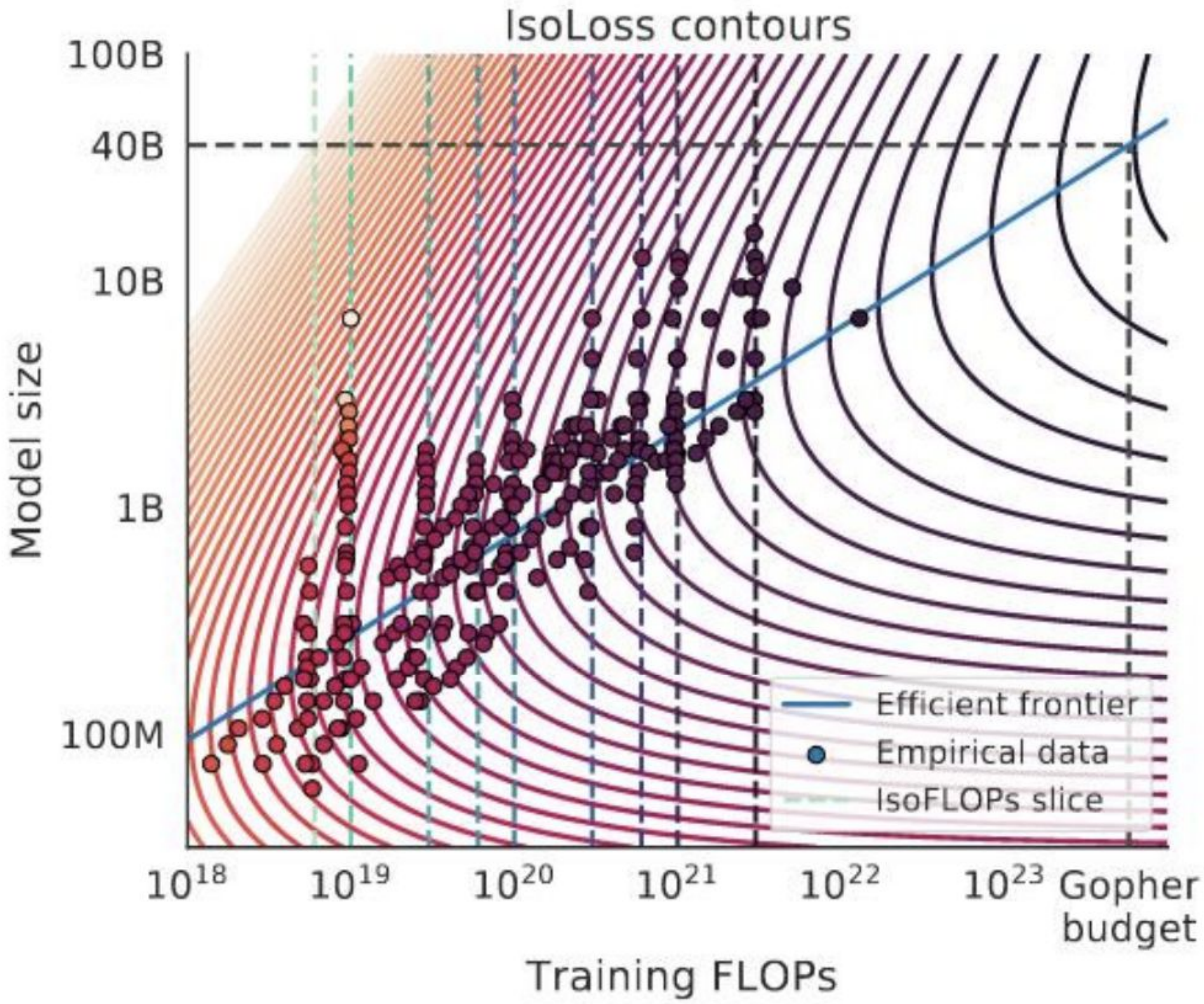
Table 2 Fits to $L(N, D)$

But Hoffman et al. (2022) tell a very different story!

Improved Scaling Laws: Hoffman et al. (2022)

Key takeaways:

- Data:
  - Fixed: C = # FLOPS
  - Experiments varied:
    - D = # tokens
    - N = # parameters
  - Measured: L(N, D)
- Learned a model to predict L(N,D) for any N and D
- Used this model to predict optimal model size



Improved Scaling Laws: Hoffman et al. (2022)

- The big shift for Chinchilla was a dramatic increase in the number of tokens
- Kaplan et al. (2020) said 8x increase in # parameters should have a 5x increase in # tokens
- But Chinchilla found you should increase both proportionally (2x # parameters + 2x # tokens)

Model	Size (# Parameters)	Training Tokens
LaMDA (Thoppilan et al., 2022)	137 Billion	168 Billion
GPT-3 (Brown et al., 2020)	175 Billion	300 Billion
Jurassic (Lieber et al., 2021)	178 Billion	300 Billion
Gopher (Rae et al., 2021)	280 Billion	300 Billion
MT-NLG 530B (Smith et al., 2022)	530 Billion	270 Billion
Chinchilla	70 Billion	1.4 Trillion

Improved Scaling Laws: Hoffman et al. (2022)

The key finding is that everyone had been using way **too little data**

Increasing the amount of data and **decreasing** the model size,
→ Same computational budget but get much better performance!

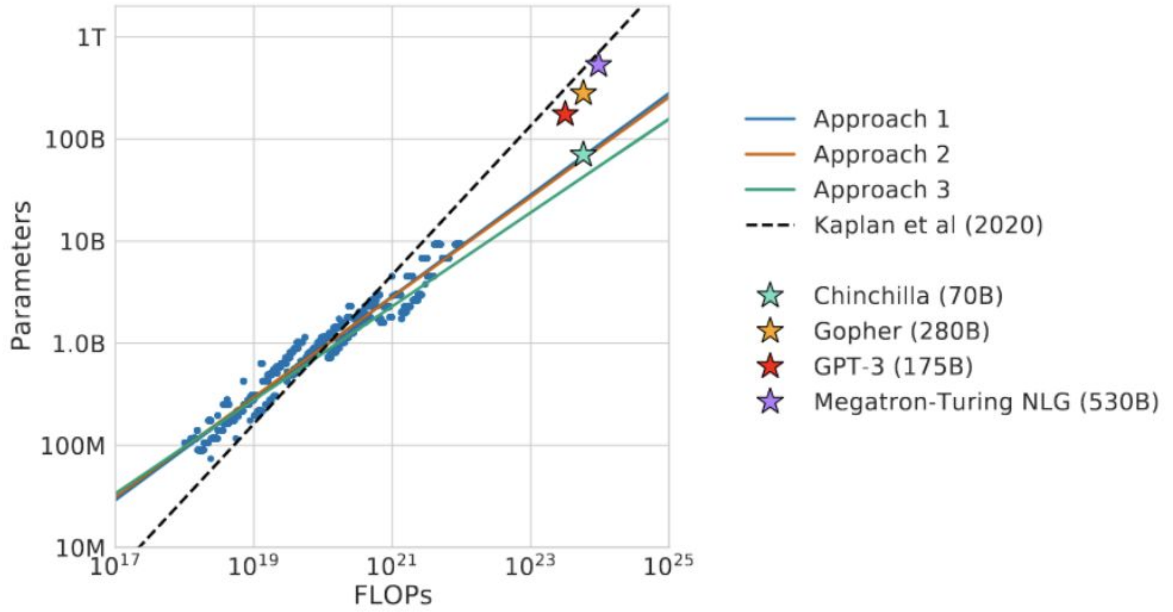


Figure 1 | **Overlaid predictions.** We overlay the predictions from our three different approaches, along with projections from Kaplan et al. (2020). We find that all three methods predict that current large models should be substantially smaller and therefore trained much longer than is currently done. In Figure A3, we show the results with the predicted optimal tokens plotted against the optimal number of parameters for fixed FLOP budgets. ***Chinchilla* outperforms *Gopher* and the other large models (see Section 4.2).**

Improved Scaling Laws: Hoffman et al. (2022)

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	Chinchilla	*Gopher*	GPT-3	MT-NLG 530B
LAMBADA Zero-Shot	**77.4**	74.5	76.2	76.6
RACE-m Few-Shot	**86.8**	75.1	58.1	-
RACE-h Few-Shot	**82.3**	71.6	46.8	47.9

Table 7 | **Reading comprehension.** On RACE-h and RACE-m (Lai et al., 2017), *Chinchilla* considerably improves performance over *Gopher*. Note that GPT-3 and MT-NLG 530B use a different prompt format than we do on RACE-h/m, so results are not comparable to *Gopher* and *Chinchilla*. On LAMBADA (Paperno et al., 2016), *Chinchilla* outperforms both *Gopher* and MT-NLG 530B.

	Chinchilla	*Gopher*	GPT-3	MT-NLG 530B	Supervised SOTA
HellaSWAG	**80.8%**	79.2%	78.9%	80.2%	93.9%
PIQA	81.8%	81.8%	81.0%	**82.0%**	90.1%
Winogrande	**74.9%**	70.1%	70.2%	73.0%	91.3%
SIQA	**51.3%**	50.6%	-	-	83.2%
BoolQ	**83.7%**	79.3%	60.5%	78.2%	91.4%

Table 8 | **Zero-shot comparison on Common Sense benchmarks.** We show a comparison between *Chinchilla*, *Gopher*, and MT-NLG 530B on various Common Sense benchmarks. We see that *Chinchilla* matches or outperforms *Gopher* and GPT-3 on all tasks. On all but one *Chinchilla* outperforms the much larger MT-NLG 530B model.

Improved Scaling Laws: Hoffman et al. (2022)

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Parameters	FLOPs	FLOPs (in *Gopher* unit)	Tokens
400 Million	1.92e+19	1/29,968	8.0 Billion
1 Billion	1.21e+20	1/4,761	20.2 Billion
10 Billion	1.23e+22	1/46	205.1 Billion
67 Billion	5.76e+23	1	1.5 Trillion
175 Billion	3.85e+24	6.7	3.7 Trillion
280 Billion	9.90e+24	17.2	5.9 Trillion
520 Billion	3.43e+25	59.5	11.0 Trillion
1 Trillion	1.27e+26	221.3	21.2 Trillion
10 Trillion	1.30e+28	22515.9	216.2 Trillion

Data Quality: Phi Family of (small) LLMs

- Key idea: Instead of increasing the size of the model / data, increase the quality of your data
- Paper Title: "Textbooks Are All You Need"
- Results from Phi-1 show performance commensurate with much larger coding models

Date	Model	Model size (Parameters)	Dataset size (Tokens)	HumanEval (Pass@1)	MBPP (Pass@1)
2021 Jul	Codex-300M [CTJ+21]	300M	100B	13.2%	-
2021 Jul	Codex-12B [CTJ+21]	12B	100B	28.8%	-
2022 Mar	CodeGen-Mono-350M [NPH+23]	350M	577B	12.8%	-
2022 Mar	CodeGen-Mono-16.1B [NPH+23]	16.1B	577B	29.3%	35.3%
2022 Apr	PaLM-Coder [CND+22]	540B	780B	35.9%	47.0%
2022 Sep	CodeGeeX [ZXZ+23]	13B	850B	22.9%	24.4%
2022 Nov	GPT-3.5 [Ope23]	175B	N.A.	47%	-
2022 Dec	SantaCoder [ALK+23]	1.1B	236B	14.0%	35.0%
2023 Mar	GPT-4 [Ope23]	N.A.	N.A.	67%	-
2023 Apr	Replit [Rep23]	2.7B	525B	21.9%	-
2023 Apr	Replit-Finetuned [Rep23]	2.7B	525B	30.5%	-
2023 May	CodeGen2-1B [NHX+23]	1B	N.A.	10.3%	-
2023 May	CodeGen2-7B [NHX+23]	7B	N.A.	19.1%	-
2023 May	StarCoder [LAZ+23]	15.5B	1T	33.6%	52.7%
2023 May	StarCoder-Prompted [LAZ+23]	15.5B	1T	40.8%	49.5%
2023 May	PaLM 2-S [ADF+23]	N.A.	N.A.	37.6%	50.0%
2023 May	CodeT5+ [WLG+23]	2B	52B	24.2%	-
2023 May	CodeT5+ [WLG+23]	16B	52B	30.9%	-
2023 May	InstructCodeT5+ [WLG+23]	16B	52B	35.0%	-
2023 Jun	WizardCoder [LXZ+23]	16B	1T	57.3%	51.8%
2023 Jun	**phi-1**	1.3B	7B	50.6%	55.5%

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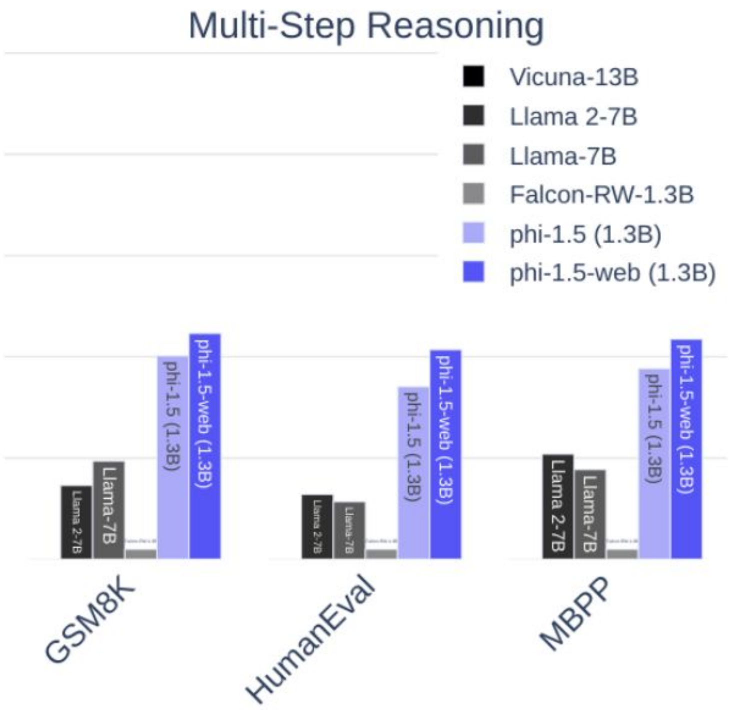
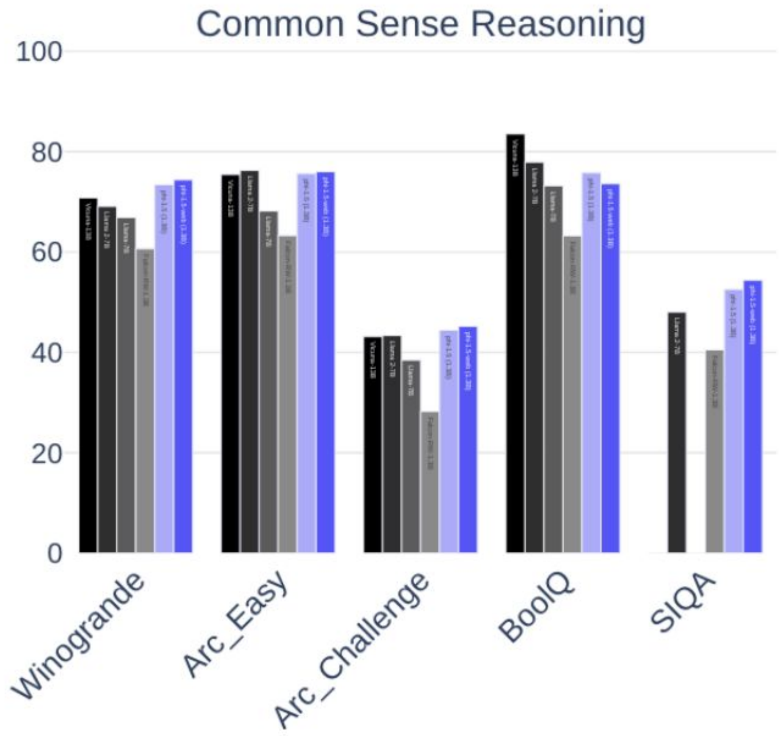
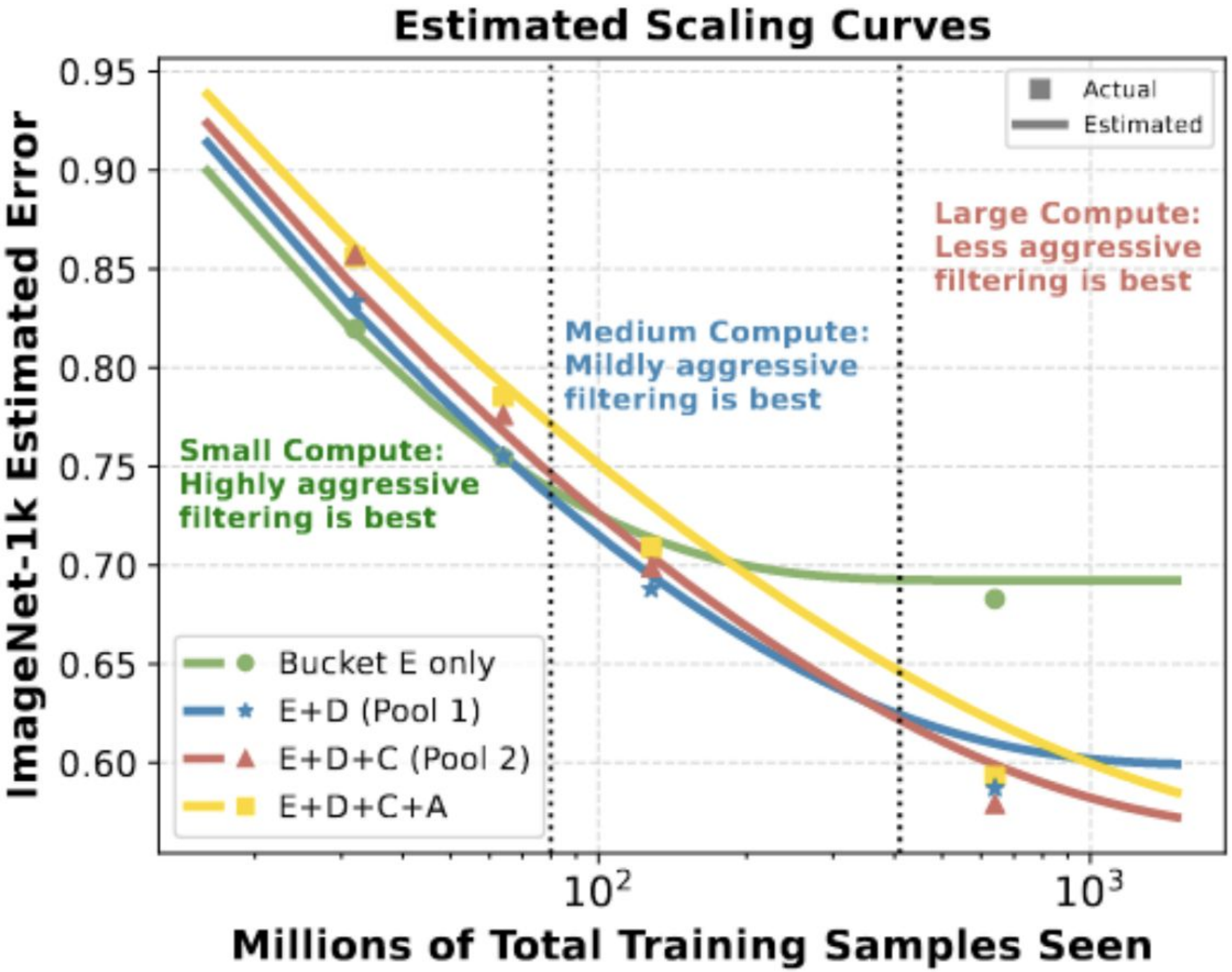


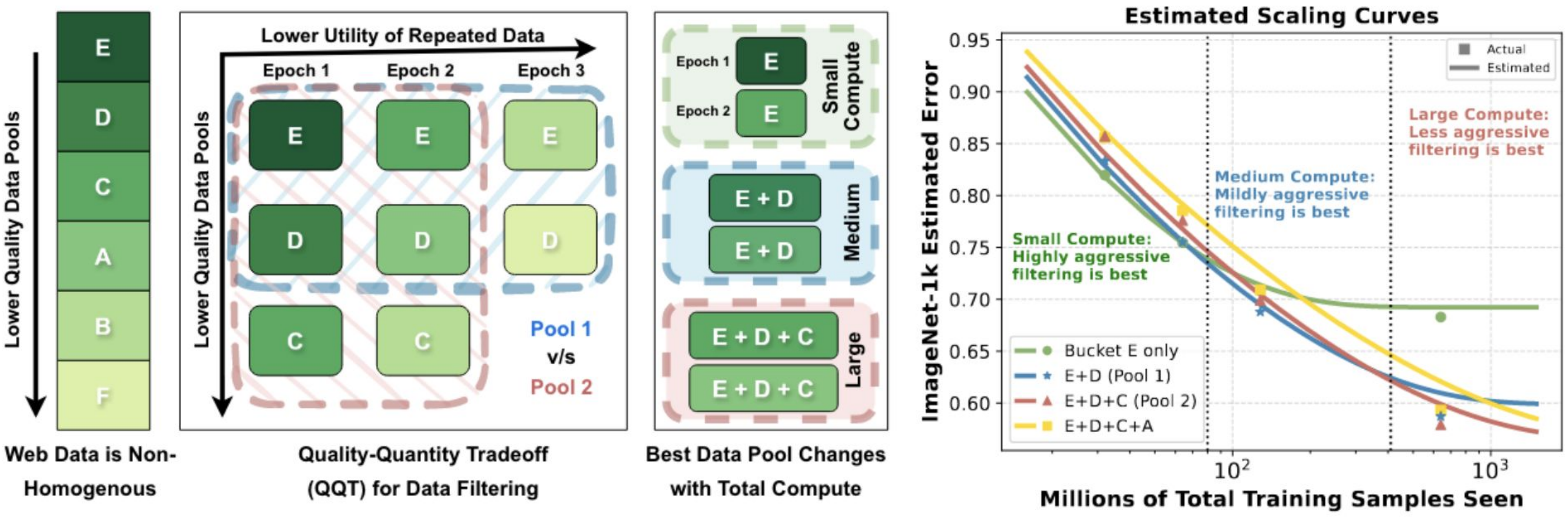


Scaling Laws for Data Filtering

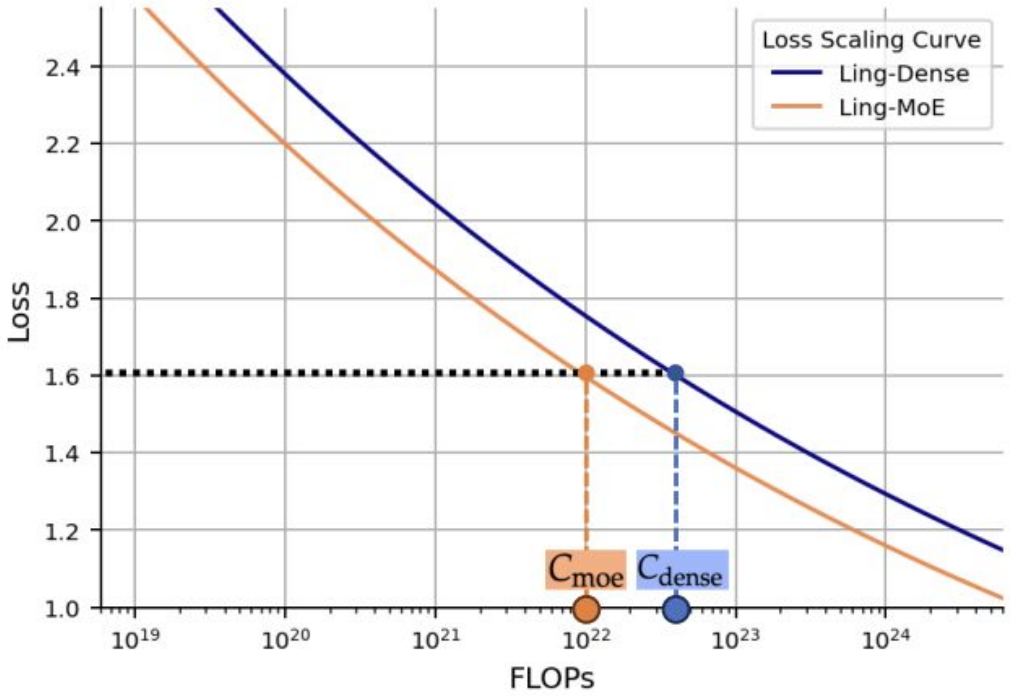
- Recent trend towards emphasizing data quality and not just data quantity
- But difficult to predict how to tradeoff between data **quantity** and **quality** and **compute**
- Scaling laws for data quality and data size (Goyal et al. 2024) suggest that as your amount of compute goes up, you can get away with less filtering



Scaling Laws for Data Filtering

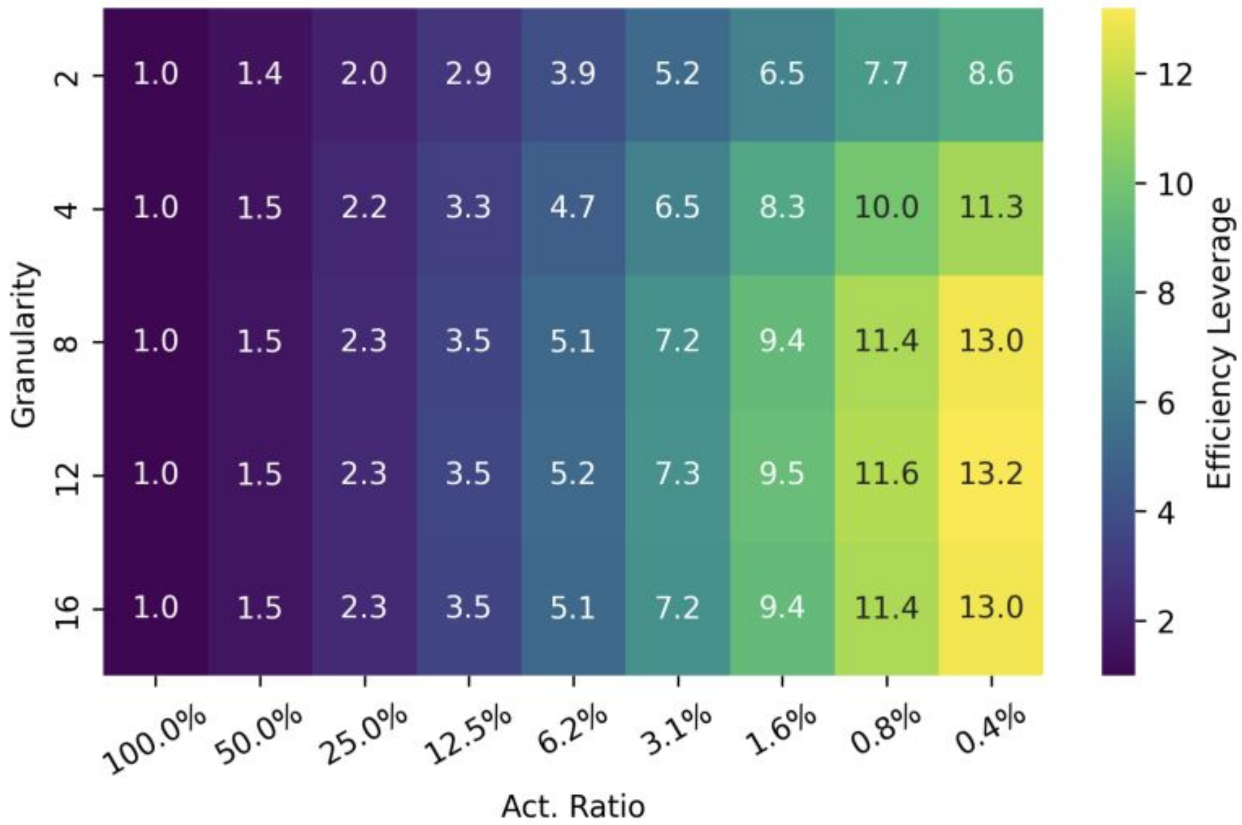


Scaling Laws for MoE



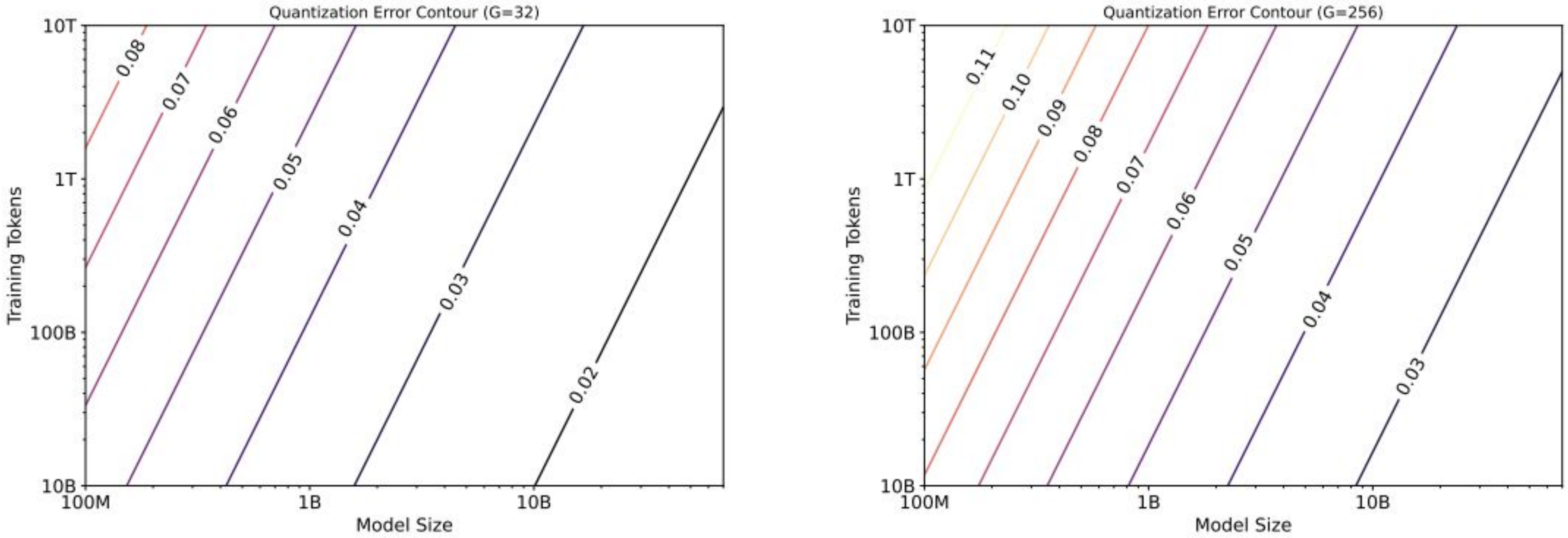
$$EL(\mathcal{X}_{\text{MoE}} \mid \mathcal{X}_{\text{Dense}}) = \frac{C_{\text{dense}}}{C_{\text{moe}}}$$

(a) Definition of Efficiency Leverage (EL)



(b) Estimated EL at $1e22$ FLOPs

Scaling Laws for Quantized Models

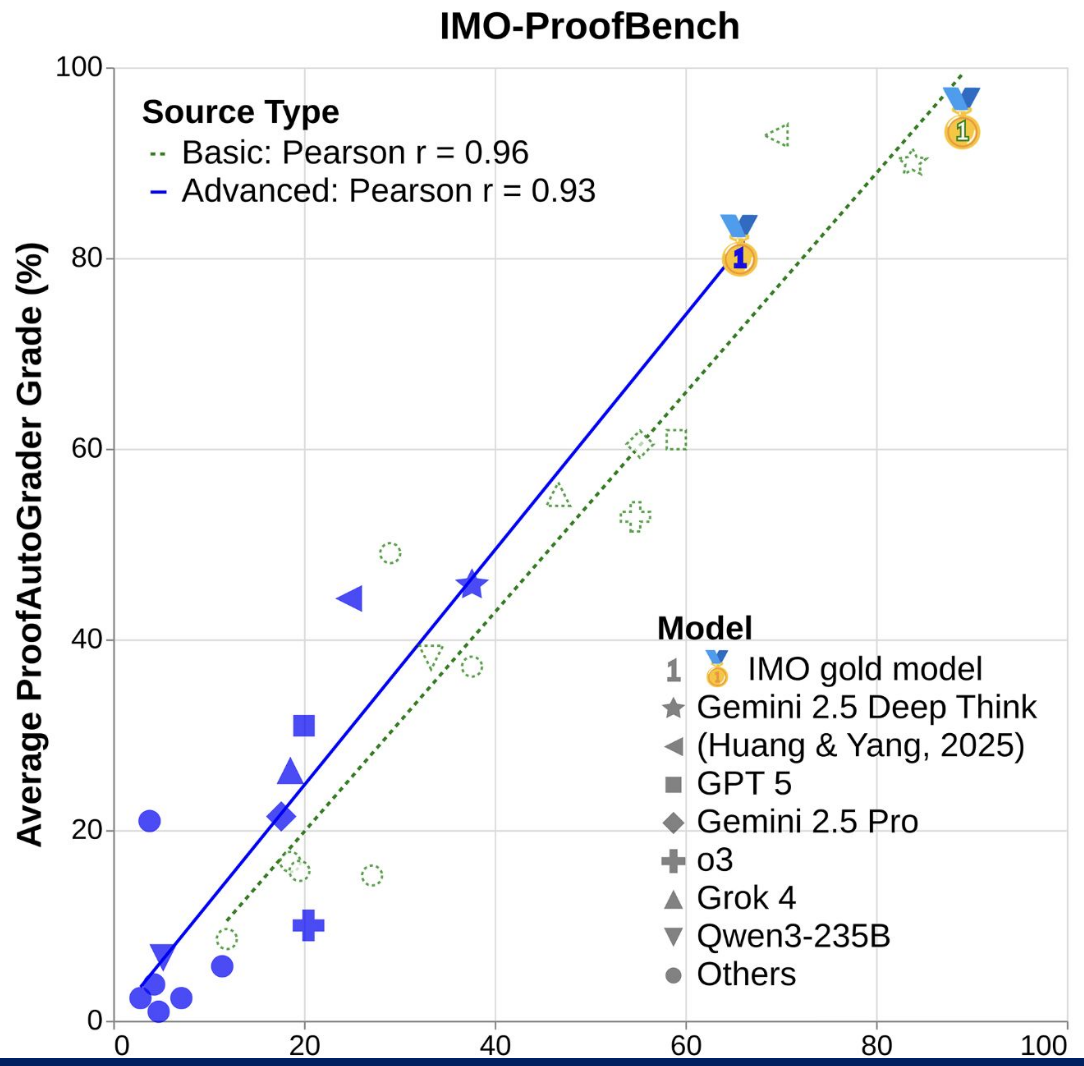


(a) W4A4G32 (b) W4A4G256

Figure 1 Quantization error contour based on the proposed unified QAT scaling law. The quantization error decreases as the model size increases, but increases with both the number of training tokens and with coarser quantization granularity.

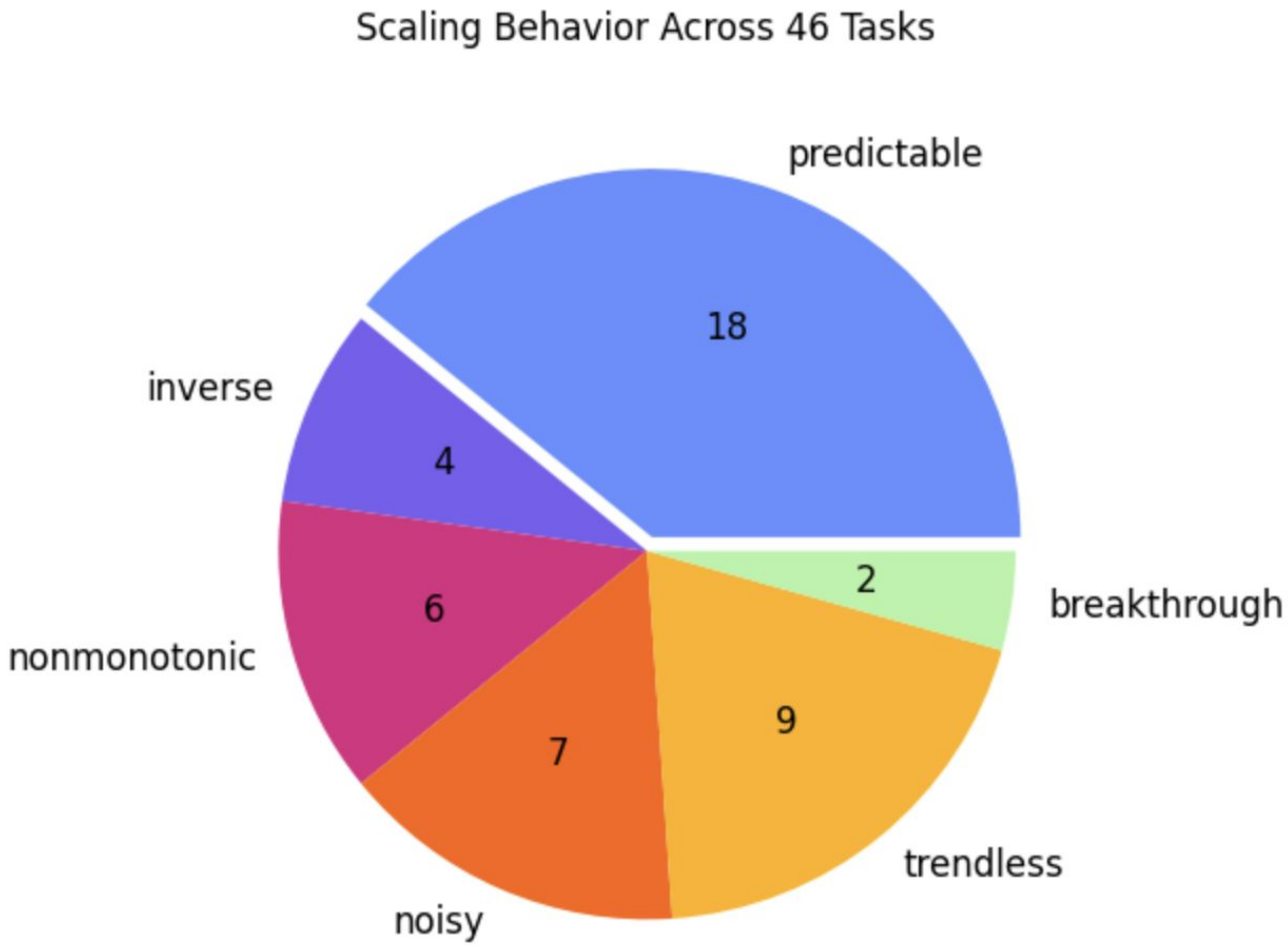
Scaling Laws Are Unreliable

Can we use scaling law to predict which model will win IMO Medal?



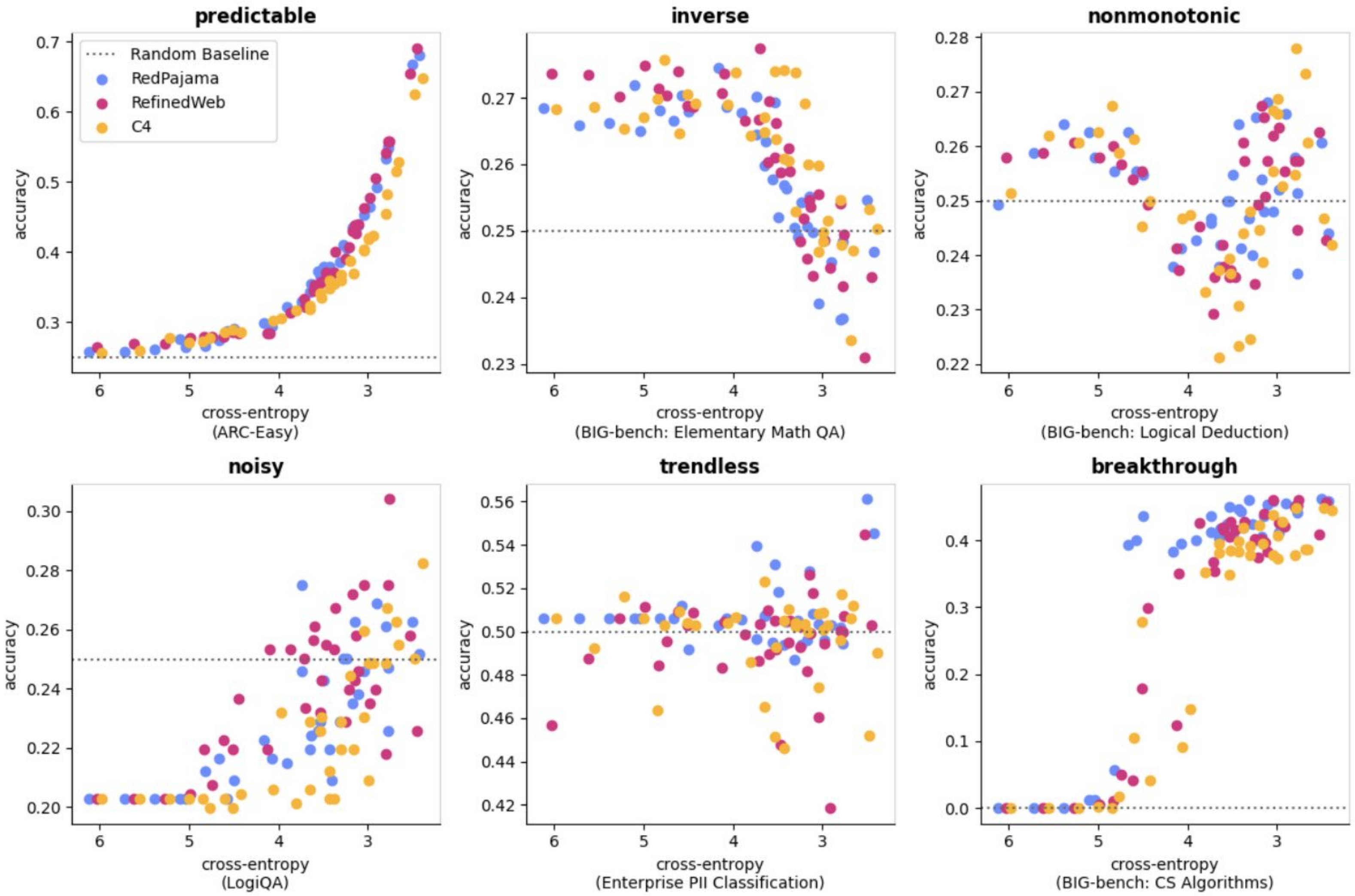
Scaling Laws Are Unreliable

Revisiting the 46 tasks and finding only 18 tasks—or 39%—demonstrate smooth, predictable improvement



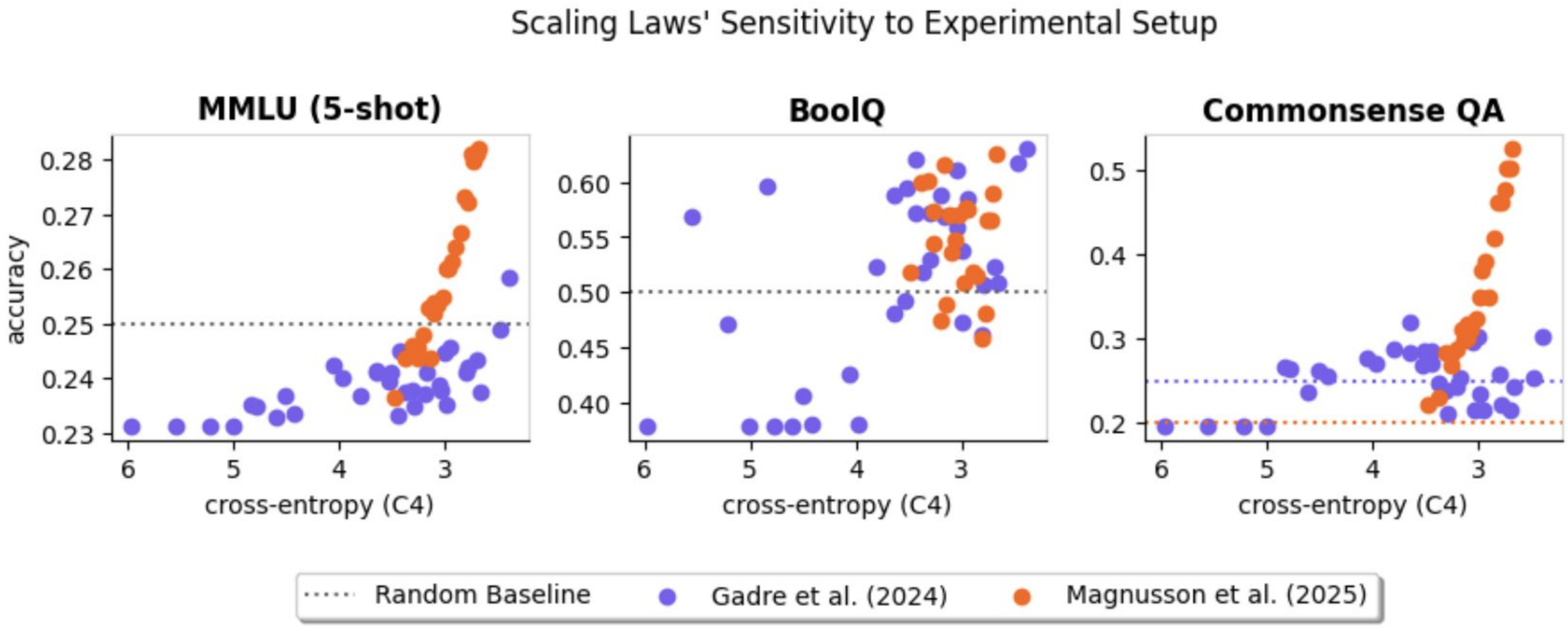
Scaling Laws Are Unreliable

The behavior of prediction results.



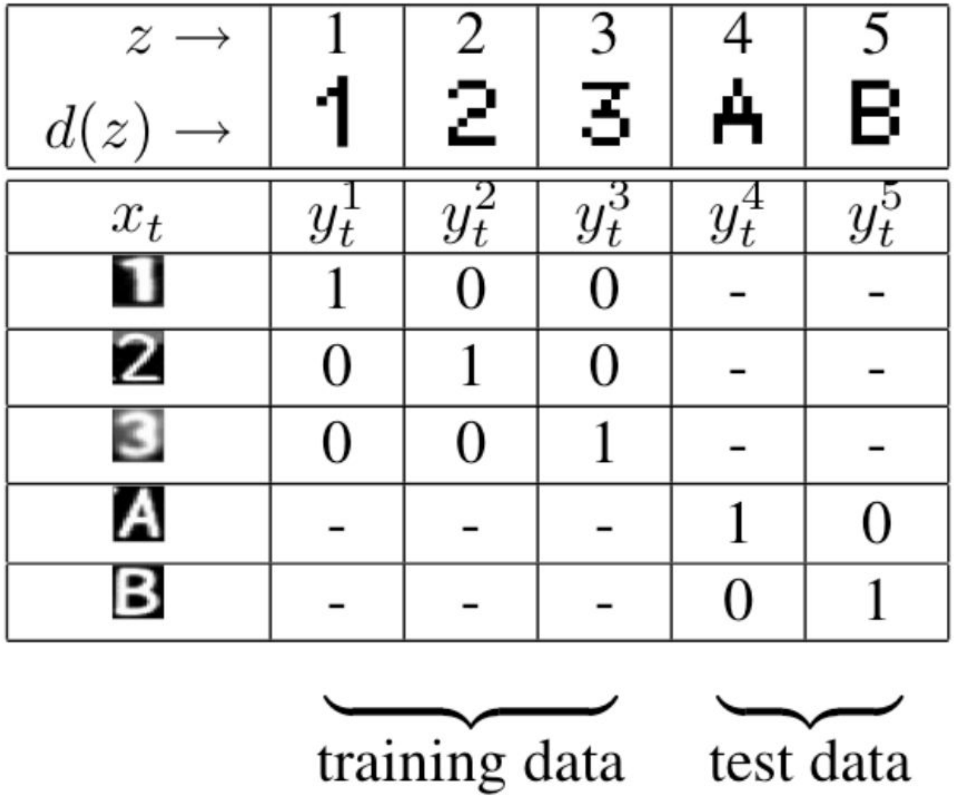
Scaling Laws Are Unreliable

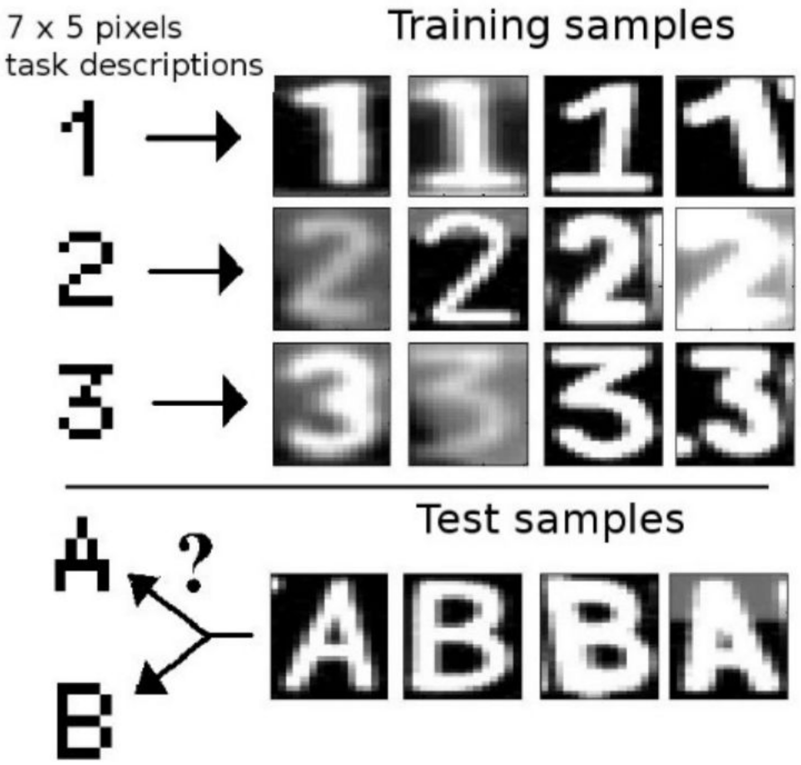
Downstream scaling laws depend on three factors: 1) pretraining data, 2) validation data, and 3) the downstream task.



Zero-shot Learning

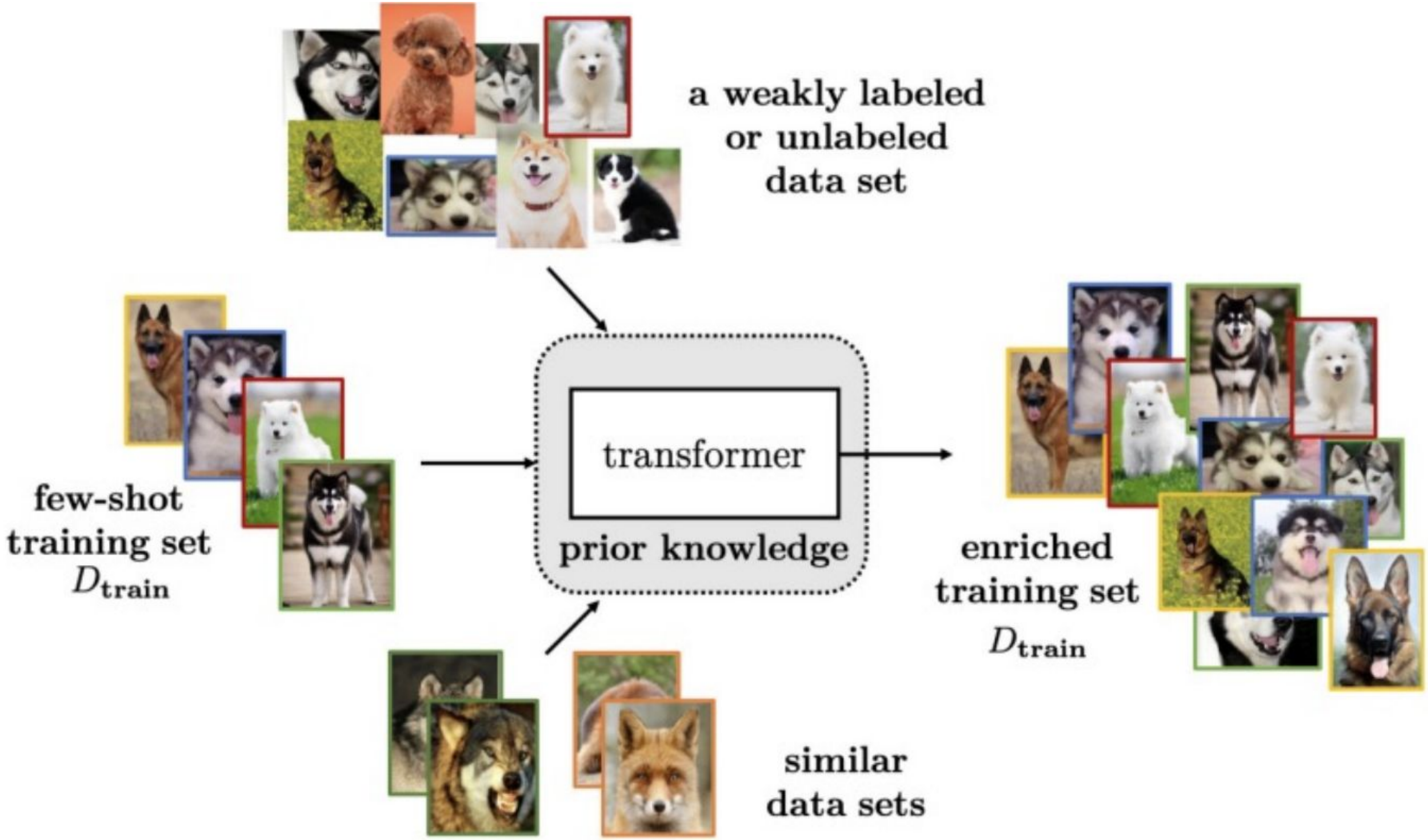
- Zero-shot learning (ZSL) is an AI technique where models classify or understand new data/tasks they weren't explicitly trained on.





Few-shot Learning

- A machine learning approach that enables a model to generalize and make accurate predictions after being trained with only a small number of labeled examples per class



Prompting

- Language models are trained to maximize the likelihood of sequences in the training data
- Most condition on the previous tokens to generate the next tokens

Definition: An *autoregressive language model* defines a probability distribution over sequences $\mathbf{x}_{1:T}$ of the form:

$$p(\mathbf{x}_{1:T}) = \prod_{t=1}^{T} p(x_t \mid x_1, \ldots, x_{t-1})$$

- Key idea behind prompting: provide a prefix string to the such that its likely completion is the answer you want

Prompting Examples

GPT-3

Prompt
(i.e. context)

```
The City
BY C. P. CAVAFY
TRANSLATED BY EDMUND KEELEY
[Poem text omitted]

SOME TREES
John Ashbery
[Poem text omitted]

Shadows on the Way
Wallace Stevens
```

Output
(i.e. generated poem)

```
I must have shadows on the way
If I am to walk I must have
Each step taken slowly and alone
To have it ready made

And I must think in lines of grey
To have dim thoughts to be my guide
Must look on blue and green
And never let my eye forget
That color is my friend
And purple must surround me too

The yellow of the sun is no more
Intrusive than the bluish snow
That falls on all of us.  I must have
Grey thoughts and blue thoughts walk with me
If I am to go away at all.
```

Conditional Sampling:

To draw a sample, feed in a prompt (i.e. context) and then sample from the model conditioned on that prompt

Prompting Examples

Llama-2-70B

Prompt

El último dueño de esta propiedad había sido un hombre soltero, que alcanzó una muy avanzada edad, y que durante gran parte de su existencia tuvo en su hermana una fiel compañera y ama de casa.

English translation:

Model Output

The last owner of this property was a bachelor man who lived to a very advanced age, and who had his sister as a faithful companion and housekeeper for most of his life.

Conditional Sampling:

To draw a sample, feed in a prompt (i.e. context) and then sample from the model conditioned on that prompt

English source from Sense & Sensibility

The late owner of this estate was a single man, who lived to a very advanced age, and who for many years of his life, had a constant companion and housekeeper in his sister.

Prompting Examples

Llama-2-70B

Prompt

Rhodochiton is a genus of flowering plants within the family Plantaginaceae, native to southern Mexico and neighbouring Guatemala. They climb by means of twining leaf stalks. One of the three species, "Rhodochiton atrosanguineus", the purple bell vine, is grown as an ornamental plant. All three species are sometimes included in "Lophospermum".

Question: You can find the purple bell vine in more than one country. True, False, or Neither?

Model Output

Answer: True.

Explanation: The purple bell vine (Rhodochiton atrosanguineus) is native to southern Mexico and neighbouring Guatemala, which means it can be found in more than one country.

Correct Answer

True.

Prompting Examples

	Llama-2-70B	Llama-2-7B Chat
Prompt	Story: Jason listened to the weather and heard it was goingto be sunny. He thought the kids might like to goswimming. He gathered up the swimsuits, towelsand sunscreen. Jason and the kids got into the truckand drove to the beach. They spent the next 2 hoursplaying and splashing in the surf. One-sentence Summary:	Story: Jason listened to the weather and heard it was goingto be sunny. He thought the kids might like to goswimming. He gathered up the swimsuits, towelsand sunscreen. Jason and the kids got into the truckand drove to the beach. They spent the next 2 hoursplaying and splashing in the surf. One-sentence Summary:
Model Output	Jason took the kids swimming at the beach after hearing the weather forecast, gathering necessary items and driving there.	Sure! Here is a one-sentence summary of the story: Jason and the kids went on a fun-filled day at the beach, playing and splashing in the surf after gathering all the necessary items, including swimsuits, towels, and sunscreen.

Story text from https://arxiv.org/pdf/1910.00998.pdf

What else can LLMs do?

Using the idea of **prompts,** we can apply LMs to a **variety of different problems** in natural language processing.

In the **zero-shot setting,** we simply feed the context to the model and observe how it completes the sequence. (i.e. there is no additional training)

Answer fact-based questions:

Context →	Organisms require energy in order to do what?
Correct Answer →	mature and develop.
Incorrect Answer →	rest soundly.
Incorrect Answer →	absorb light.
Incorrect Answer →	take in nutrients.

Complete sentences logically:

Context →	My body cast a shadow over the grass because
Correct Answer →	the sun was rising.
Incorrect Answer →	the grass was cut.

Complete analogies:

Context →	lull is to trust as
Correct Answer →	cajole is to compliance
Incorrect Answer →	balk is to fortitude
Incorrect Answer →	betray is to loyalty
Incorrect Answer →	hinder is to destination
Incorrect Answer →	soothe is to passion

Reading comprehension:

Context →	anli 1: anli 1: Fulton James MacGregor MSP is a Scottish politician who is a Scottish National Party (SNP) Member of Scottish Parliament for the constituency of Coatbridge and Chryston. MacGregor is currently Parliamentary Liaison Officer to Shona Robison, Cabinet Secretary for Health & Sport. He also serves on the Justice and Education & Skills committees in the Scottish Parliament. Question: Fulton James MacGregor is a Scottish politican who is a Liaison officer to Shona Robison who he swears is his best friend. True, False, or Neither?
Correct Answer →	Neither
Incorrect Answer →	True
Incorrect Answer →	False

Examples from GPT-3

Zero -shot LLMs

- GPT-2 (1.5B parameters) for unsupervised prediction on various tasks
- GPT-2 models p(output | input, task)
  - translation: *(translate to french, english text, french text)*
  - reading comprehension: *(answer the question, document, question, answer)*
- Why does this work?

"I'm not the cleverest man in the world, but like they say in French: **Je ne suis pas un imbecile [I'm not a fool].**

In a now-deleted post from Aug. 16, Soheil Eid, Tory candidate in the riding of Joliette, wrote in French: "**Mentez mentez, il en restera toujours quelque chose,**" which translates as, "**Lie lie and something will always remain.**"

"I hate the word '**perfume**,"' Burr says. 'It's somewhat better in French: '**parfum**.'

If listened carefully at 29:55, a conversation can be heard between two guys in French: "**-Comment on fait pour aller de l'autre coté? -Quel autre coté?**", which means "**- How do you get to the other side? - What side?**".

If this sounds like a bit of a stretch, consider this question in French: **As-tu aller au cinéma?**, or **Did you go to the movies?**, which literally translates as Have-you to go to movies/theater?

"**Brevet Sans Garantie Du Gouvernement**", translated to English: "**Patented without government warranty**".

Table 1. Examples of naturally occurring demonstrations of English to French and French to English translation found throughout the WebText training set.

Zero -shot LLMs

- GPT-2 (1.5B parameters) for unsupervised prediction on various tasks
- GPT-2 models p(output | input, task)
  - translation: *(translate to french, english text, french text)*
  - reading comprehension: *(answer the question, document, question, answer)*
- Why does this work?

Language Models are Unsupervised Multitask Learners

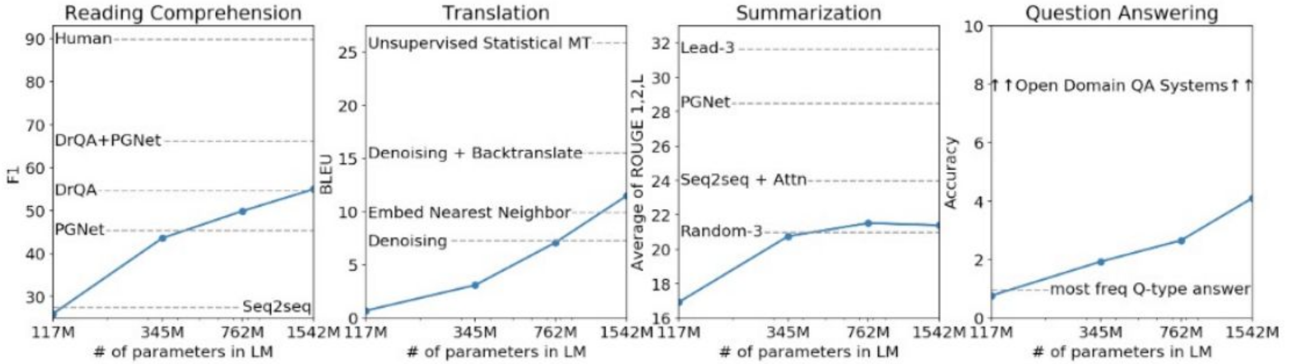


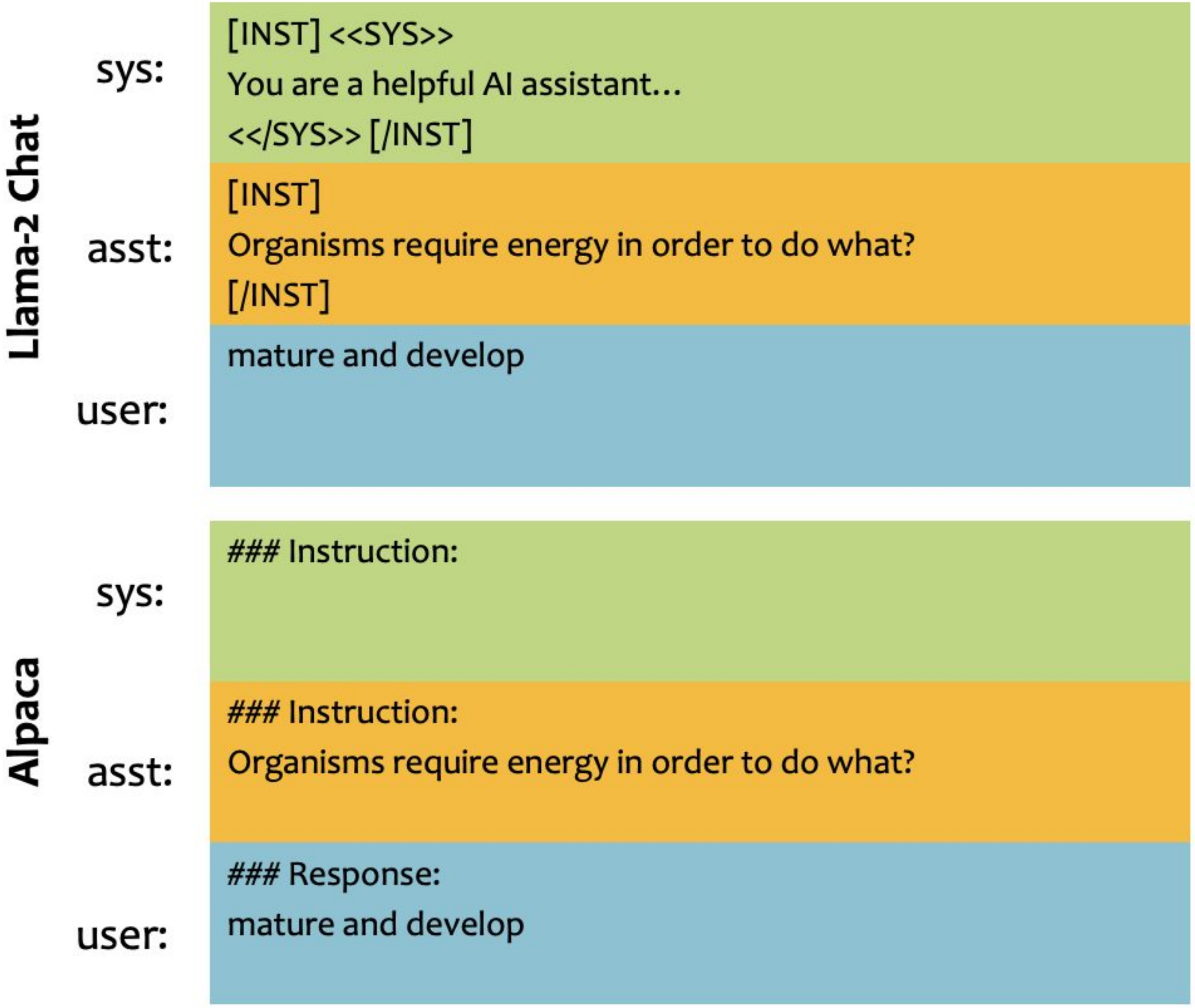
Figure 1. Zero-shot task performance of WebText LMs as a function of model size on many NLP tasks. Reading Comprehension results are on CoQA (Reddy et al., 2018), translation on WMT-14 Fr-En (Artetxe et al., 2017), summarization on CNN and Daily Mail (See et al., 2017), and Question Answering on Natural Questions (Kwiatkowski et al., 2019). Section 3 contains detailed descriptions of each result.

	LAMBADA (PPL)	LAMBADA (ACC)	CBT-CN (ACC)	CBT-NE (ACC)	WikiText2 (PPL)	PTB (PPL)	enwik8 (BPB)	text8 (BPC)	WikiText103 (PPL)	1BW (PPL)
SOTA	99.8	59.23	85.7	82.3	39.14	46.54	0.99	1.08	18.3	**21.8**
117M	**35.13**	45.99	**87.65**	**83.4**	**29.41**	65.85	1.16	1.17	37.50	75.20
345M	**15.60**	55.48	**92.35**	**87.1**	**22.76**	47.33	1.01	**1.06**	26.37	55.72
762M	**10.87**	**60.12**	**93.45**	**88.0**	**19.93**	**40.31**	**0.97**	**1.02**	22.05	44.575
1542M	**8.63**	**63.24**	**93.30**	**89.05**	**18.34**	**35.76**	**0.93**	**0.98**	**17.48**	42.16

Table 3. Zero-shot results on many datasets. No training or fine-tuning was performed for any of these results. PTB and WikiText-2 results are from (Gong et al., 2018). CBT results are from (Bajgar et al., 2016). LAMBADA accuracy result is from (Hoang et al., 2018) and LAMBADA perplexity result is from (Grave et al., 2016). Other results are from (Dai et al., 2019).

Zero -shot LLMs

- Models like ChatGPT, Llama-2 Chat, etc. have been fine-tuned as chat assistants
- These (often) were trained with specific prompt templates that segment the prompt into different parts: (1) system (2) assistant (3) user



Prompting for Instruction Fine-tuned Models

Prompt

Model Output

Llama-2-70B

Prompt:

Story: Jason listened to the weather
and heard it was goingto be
thought the kids might like
goswimming. He gathered u
swimsuits, towelsand sunsc
Jason and the kids got into
truckand drove to the beach
spent the next 2 hoursplayi
splashing in the surf.

One-sentence Summary:

Model Output:

Jason took the kids swimming at the beach after hearing the weather forecast, gathering necessary items and driving there.

Lllama-2-7B Chat was *instruction fine-tuned* (more on this later) and so its responses look quite different from the Llama-2 models that were not

Llama-2-7B Chat

Prompt:

Story: Jason listened to the weather
t was goingto be sunny. He
kids might like to
g. He gathered up the
owelsand sunscreen.
he kids got into the
ove to the beach. They
ext 2 hoursplaying and
the surf.

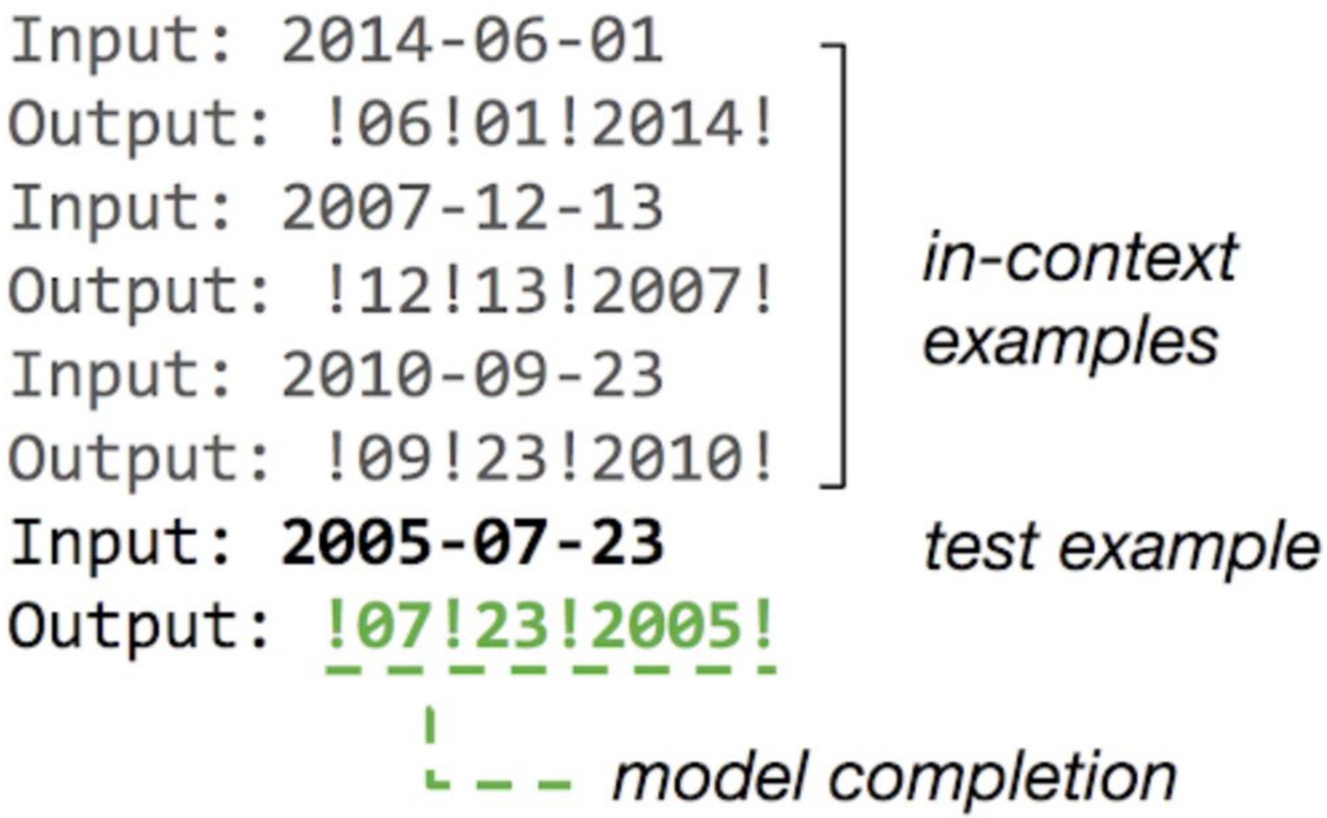
One-sentence Summary:

Model Output:

Sure! Here is a one-sentence summary of the story:
Jason and the kids went on a fun-filled day at the beach, playing and splashing in the surf after gathering all the necessary items, including swimsuits, towels, and sunscreen.

Story text from https://arxiv.org/pdf/1910.00998.pdf

In-context Learning



Few-shot Learning with LLMs

Suppose you have…

- a dataset $D = \{(x_i, y_i)\}_{i=1}^{N}$ and N is rather small (i.e. few-shot setting)
- a very large (billions of parameters) pre-trained language model

There are two ways to "learn"

This section!

Option A: Supervised fine-tuning

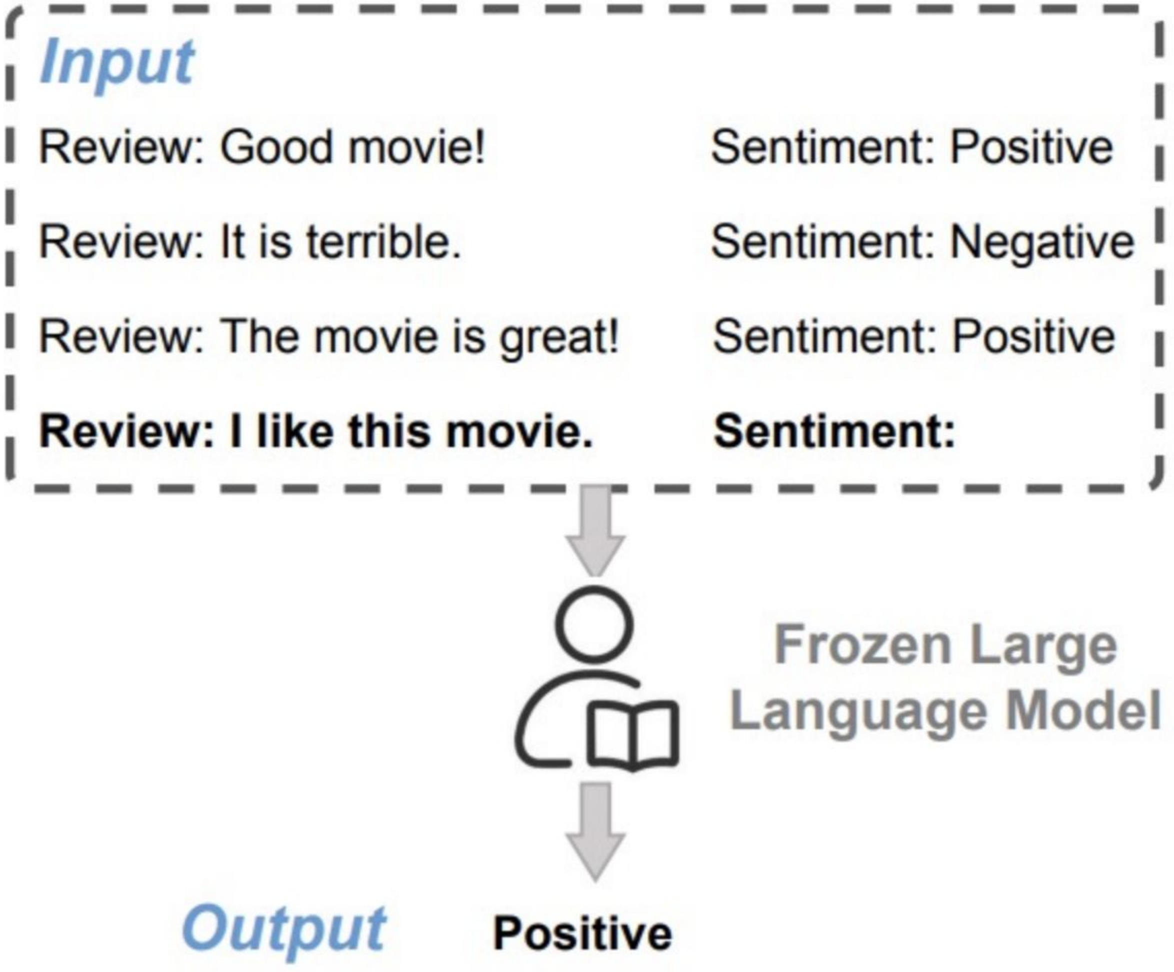
- **Definition:** fine-tune the LLM on the training data using…
  - a standard supervised objective
  - backpropagation to compute gradients
  - your favorite optimizer (e.g. Adam)
- **Pro:** fits into the standard ML recipe
- **Pro:** still works if N is large
- **Con:** backpropagation requires ~3x the memory and computation time as the forward computation
- **Con:** you might not have access to the model weights at all (e.g. because the model is proprietary)

Option B: In-context learning

- **Definition:**
  1. feed training examples to the LLM as a prompt
  2. allow the LLM to infer patterns in the training examples during inference (i.e. decoding)
  3. take the output of the LLM following the prompt as its prediction
- **Con:** the prompt may be very long and Transformer LMs require $O(N^2)$ time/space where N = length of context
- **Pro:** no backpropagation required and only one pass through the training data
- **Pro:** does not require model weights, only API access

Few-shot In-context Learning

- Few-shot learning can be done via in-context learning
- Typically, a task description is presented first
- Then a sequence of input/output pairs from a training dataset are presented in sequence



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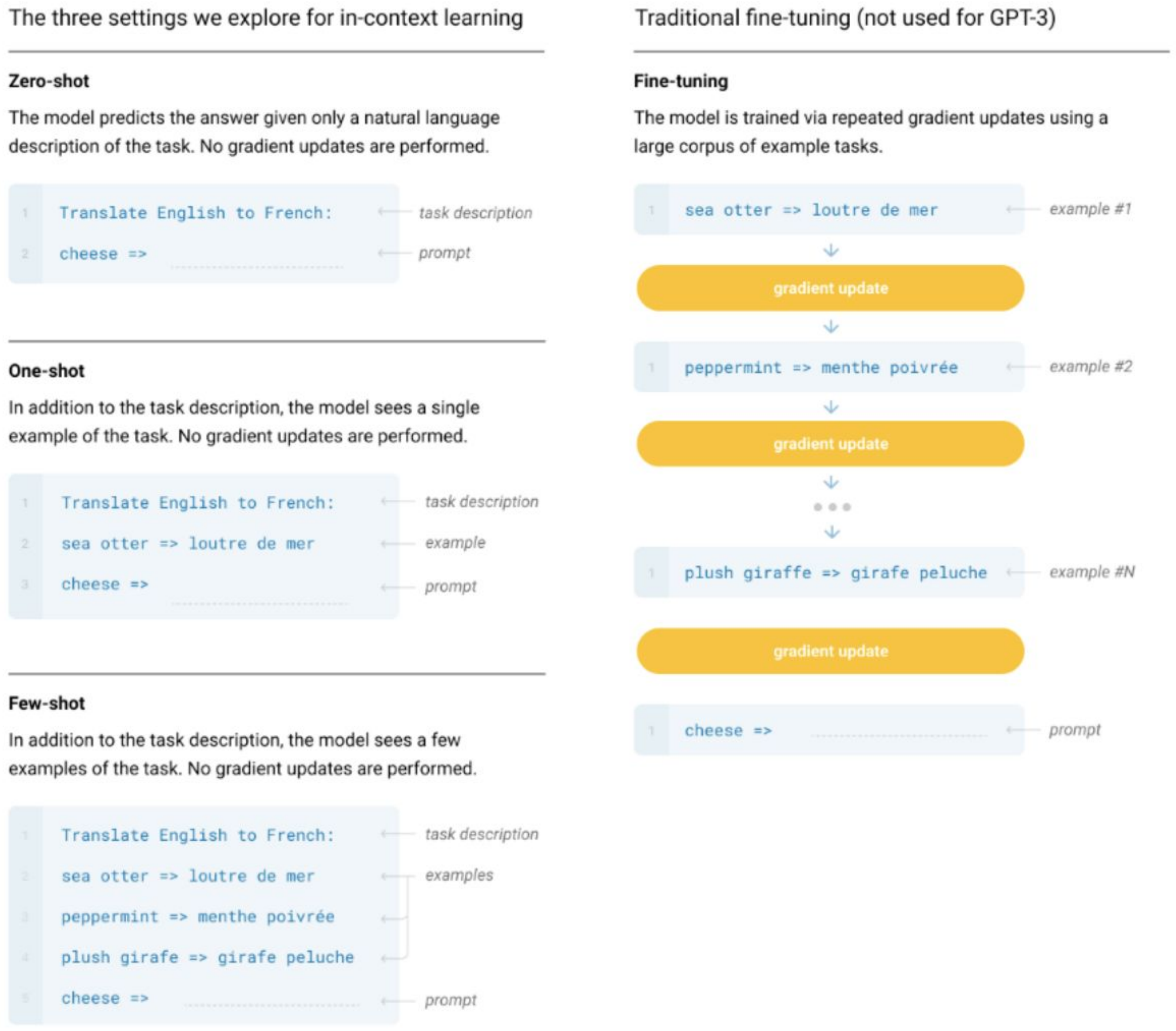


Few-shot In-context Learning

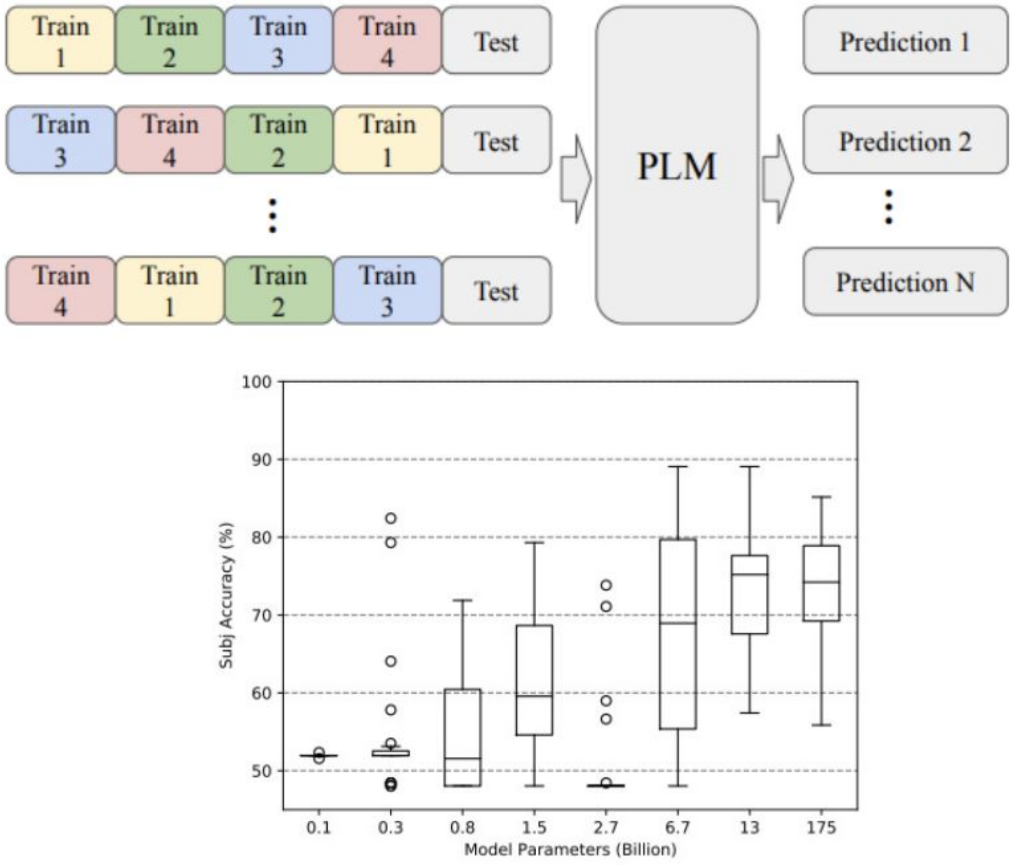
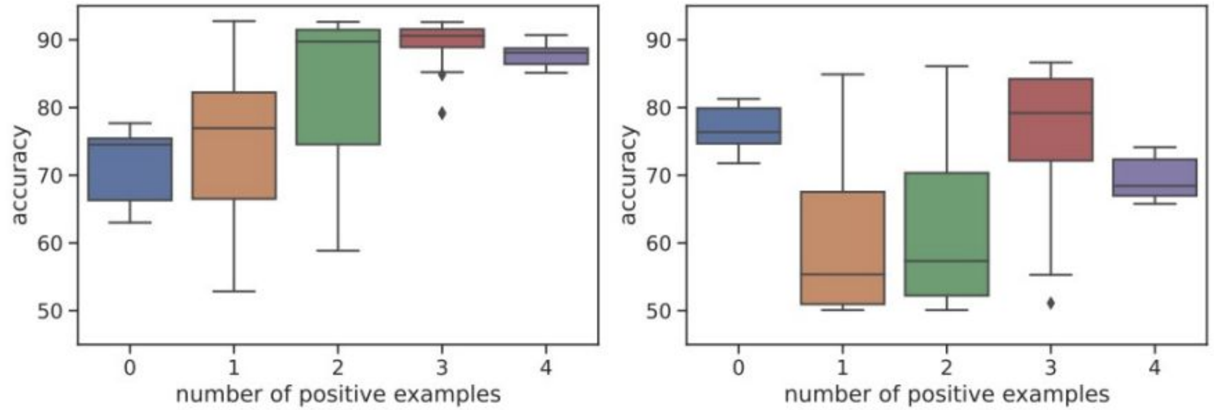


Figure 1: Four-shot performance for 24 different sample orders across different sizes of GPT-family models (GPT-2 and GPT-3) for the SST-2 and Subj datasets.

In-context learning can be sensitive to…

1. **the order the training examples are presented**
2. the balance of labels (e.g. positive vs. negative)
3. the number of unique labels covered

Few-shot In-context Learning



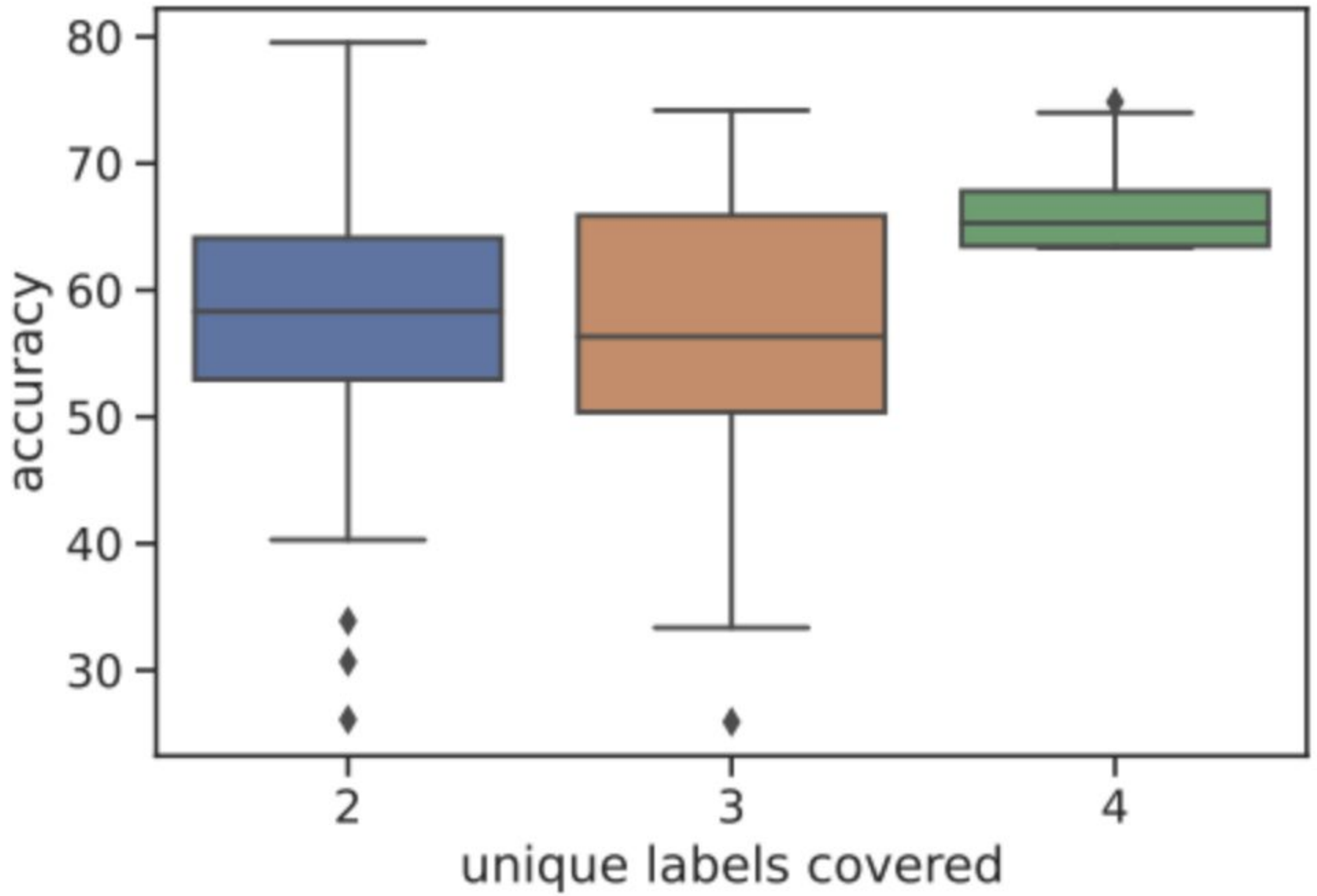
(a) Amazon (b) SST-2

Figure 3: Accuracies of Amazon and SST-2 with varying **label balance** (number of positive examples in demonstration), across 100 total random samples of 4 demonstration examples.

In-context learning can be sensitive to…

1. the order the training examples are presented
2. **the balance of labels (e.g. positive vs. negative)**
3. the number of unique labels covered

Few-shot In-context Learning



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3. **the number of unique labels covered**

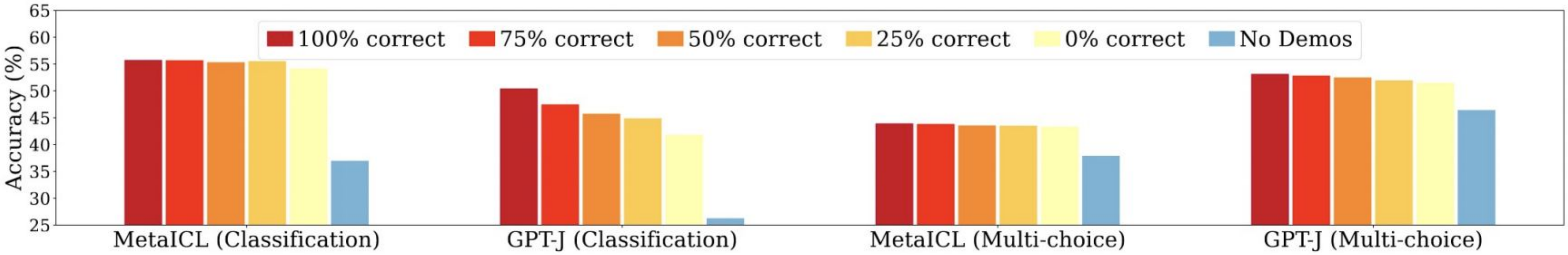
Few-shot In-context Learning

You would expect these to be important…

A. whether or not the training examples have the true label (as opposed to a random one)

B. having more in-context training examples

…but it's not always the case



Few-shot In-context Learning

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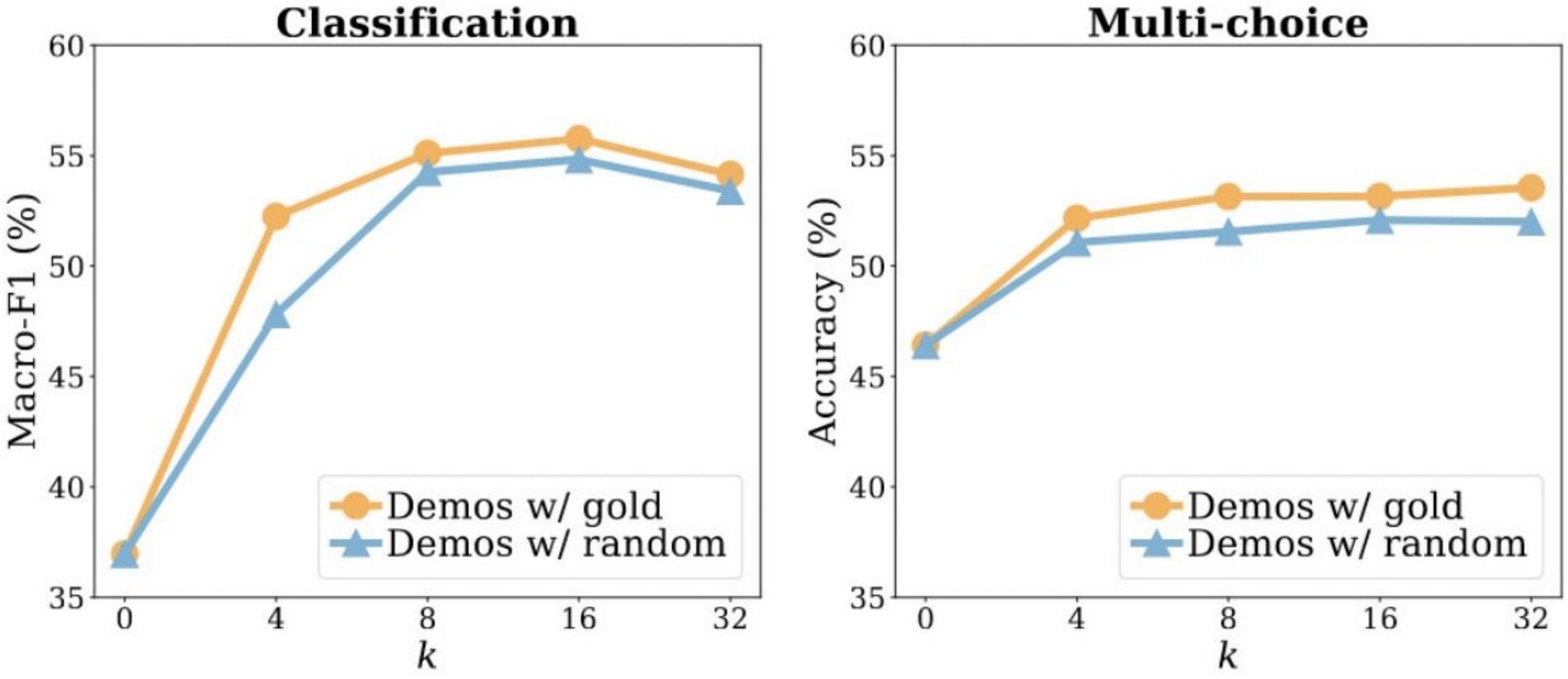
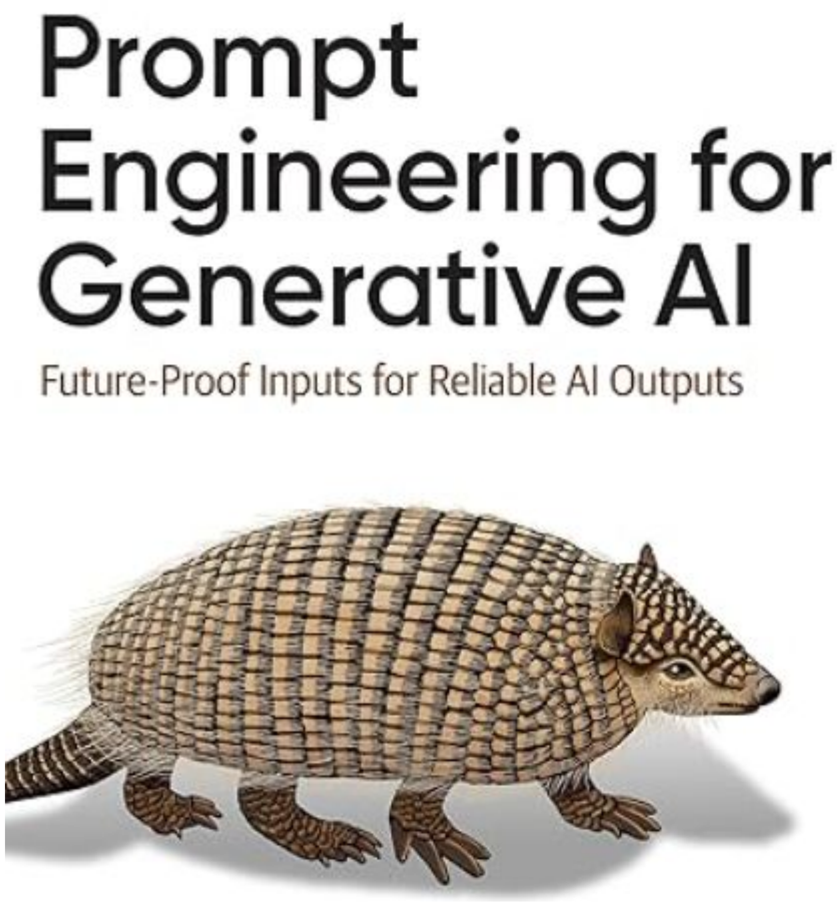


Prompt Engineering



Prompt Engineering

- **Task:** News topic classification
- **Dataset:** AG News
- **Model:** OPT-175B
- **Setup:** zero-shot learning

Question: if we evaluate the model multiple times keeping everything fixed except for the prompt, do we always get the same results?

Prompt	**Accuracy**
What is this piece of news regarding?	40.9
What is this article about?	52.4
What is the best way to describe this article?	68.2
What is the most accurate label for this news article?	71.2

Prompt Engineering

- **Task:** News topic classification
- **Dataset:** AG News
- **Model:** OPT-175B
- **Setup:** zero-shot learning

Question: how can we pick a good prompt?

Answer: pick the prompt with the lowest perplexity under the model!

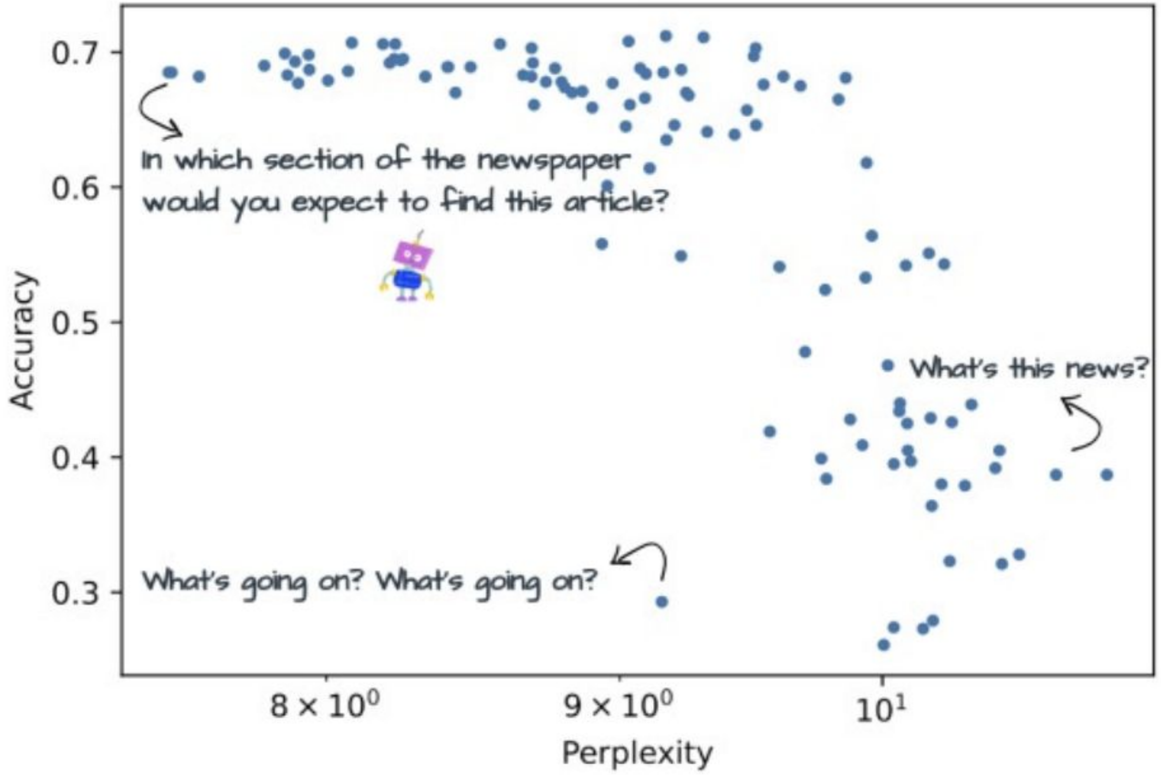


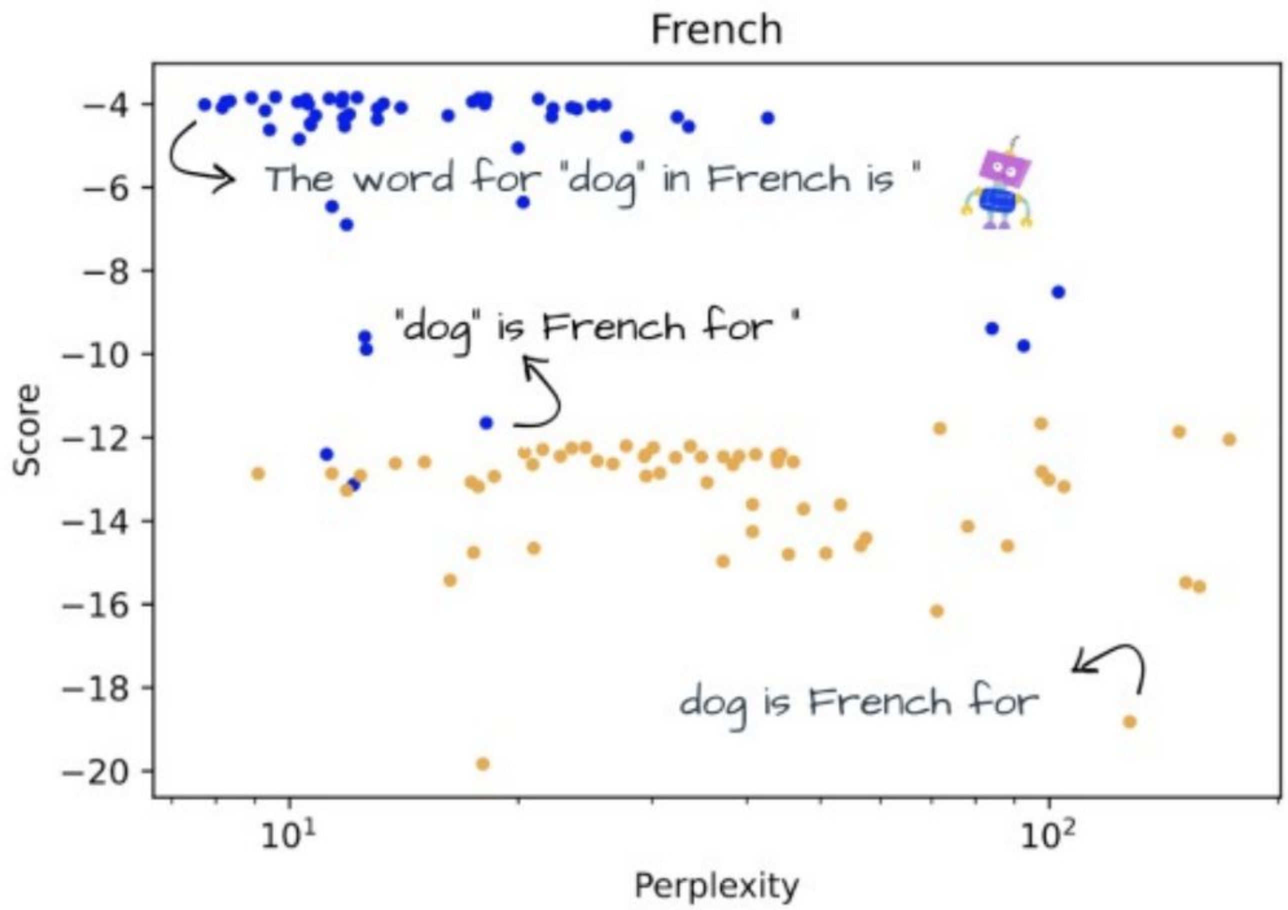
Figure 1: Accuracy vs. perplexity for the AG News dataset with OPT 175b. The x axis is in log scale. Each point stands for a different prompt.

Prompt Engineering

- **Task:** French word-level translation
- **Dataset:** NorthEuraLex
- **Model:** Bloom (multilingual LLM)
- **Setup:** zero-shot learning

Question: how can we pick a good prompt?

Answer: pick the prompt with the lowest perplexity under the model!



Prompt Engineering - Resources

prompts.chat

The world's largest open-source prompt library for AI

Works with ChatGPT, Claude, Gemini, Llama, Mistral, and more

Website prompts.chat | awesome | 🤗 Hugging Face | Ask DeepWiki

🌐 Browse Prompts • 📖 Read the Book • 📄 View on GitHub • 🚀 Self-Host

🏆 Featured in Forbes · 🎓 Referenced by Harvard, Columbia · 📄 40+ academic citations · ❤️ Most liked dataset on Hugging Face

⭐ 143k+ GitHub stars · 🏅 GitHub Staff Pick · 🚀 First prompt library (Dec 2022)

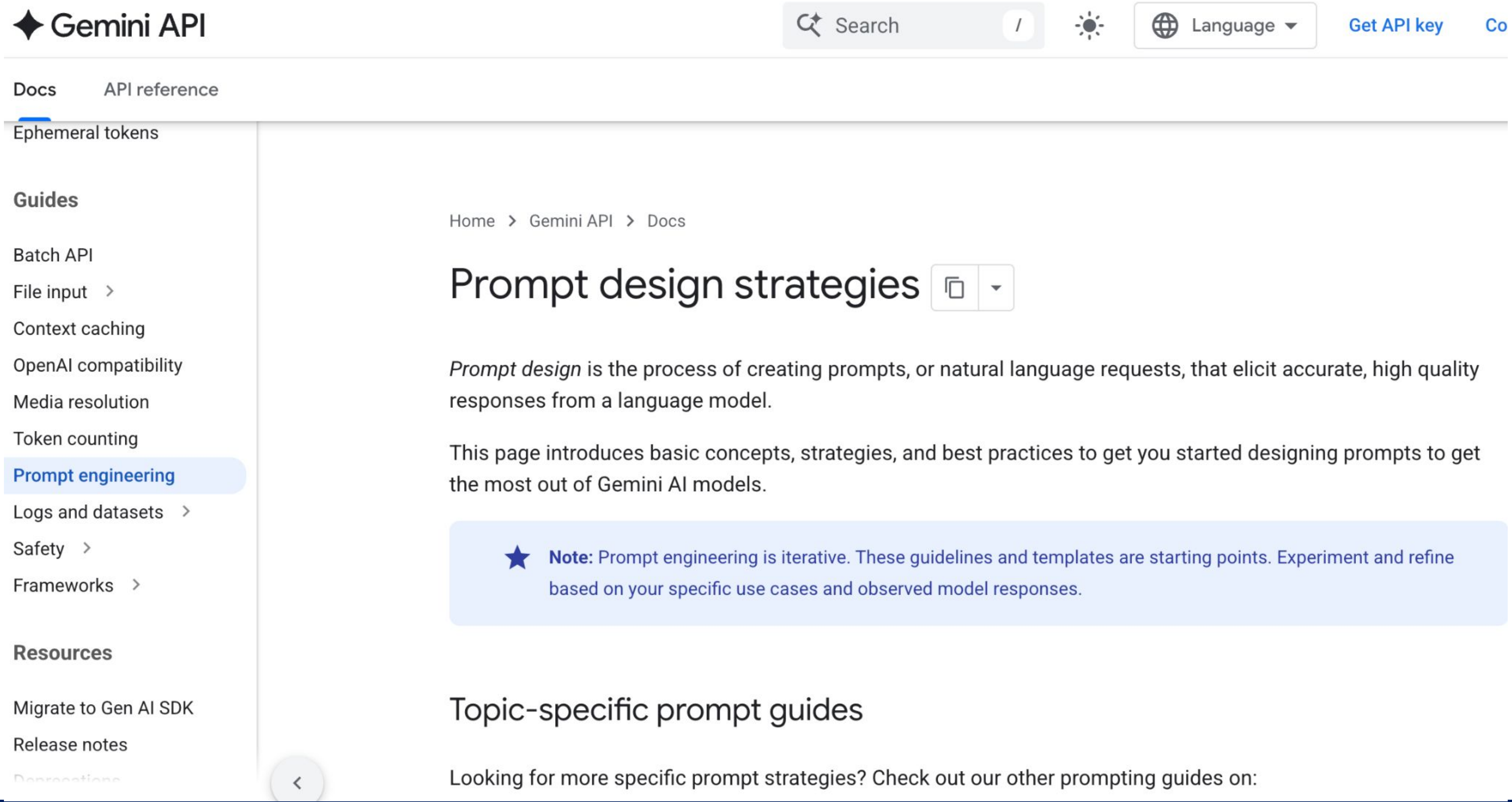
Loved by AI pioneers:

Greg Brockman (OpenAI Co-Founder) · **Wojciech Zaremba** (OpenAI Co-Founder) · **Clement Delangue** (Hugging Face CEO) · **Thomas Dohmke** (Former GitHub CEO)

What is this?

A curated collection of **prompt examples** for AI chat models. Originally created for ChatGPT, these prompts work

Prompt Engineering - Resources



Gemini API

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Ephemeral tokens

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Media resolution
Token counting
Prompt engineering
Logs and datasets
Safety
Frameworks

Resources

Migrate to Gen AI SDK
Release notes

Home > Gemini API > Docs

Prompt design strategies

Prompt design is the process of creating prompts, or natural language requests, that elicit accurate, high quality responses from a language model.

This page introduces basic concepts, strategies, and best practices to get you started designing prompts to get the most out of Gemini AI models.

Note: Prompt engineering is iterative. These guidelines and templates are starting points. Experiment and refine based on your specific use cases and observed model responses.

Topic-specific prompt guides

Looking for more specific prompt strategies? Check out our other prompting guides on:

Prompt Engineering - Resources

🧠 Awesome Claude Prompts

awesome License MIT

- Author: yzfly / 云中江树

Welcome to the "Awesome Claude Prompts" repository! This is a collection of prompt examples to be used with the Claude model.

The Claude model is an AI assistant created by Anthropic that is capable of generating human-like text. By providing it with a prompt, it can generate responses that continue the conversation or expand on the given prompt.

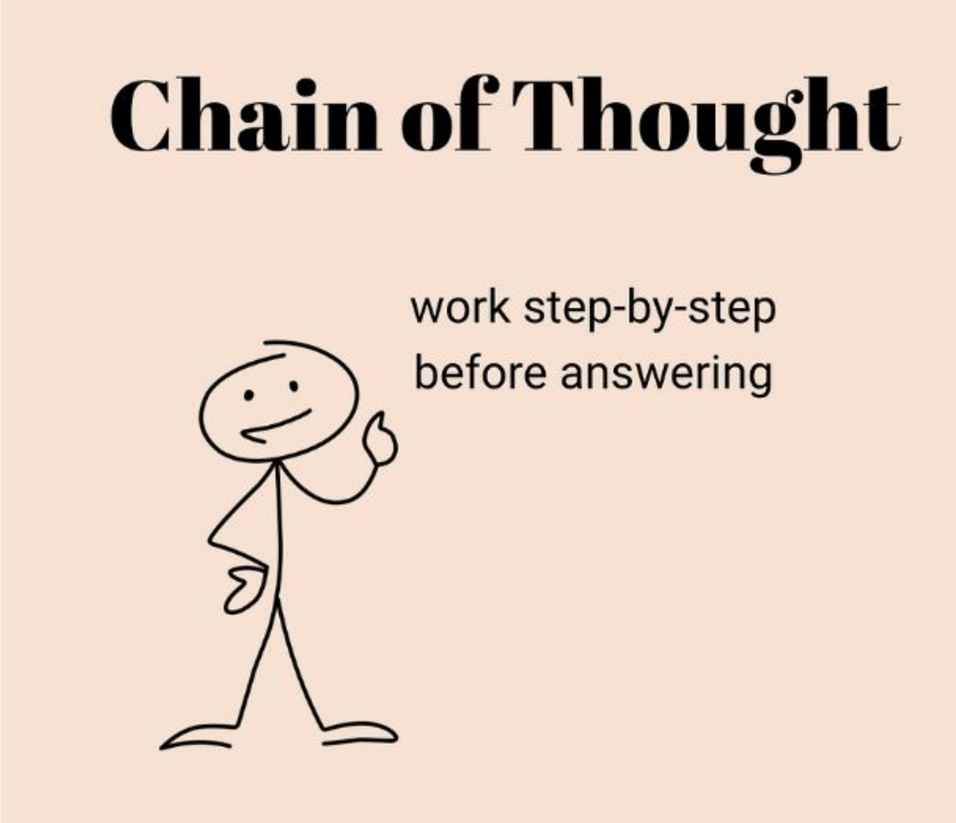
Claude offers many amazing features that ChatGPT does not support, such as longer contexts (up to 100k), free file uploading, etc., making it more powerful than ChatGPT.

In this repository, you will find a variety of prompts that can be used with Claude. We encourage you to add your own prompts to the list, and to use Claude to generate new prompts as well.

To get started, simply clone this repository and use the prompts in the README.md file as input for Claude. You can also use the prompts in this file as inspiration for creating your own.

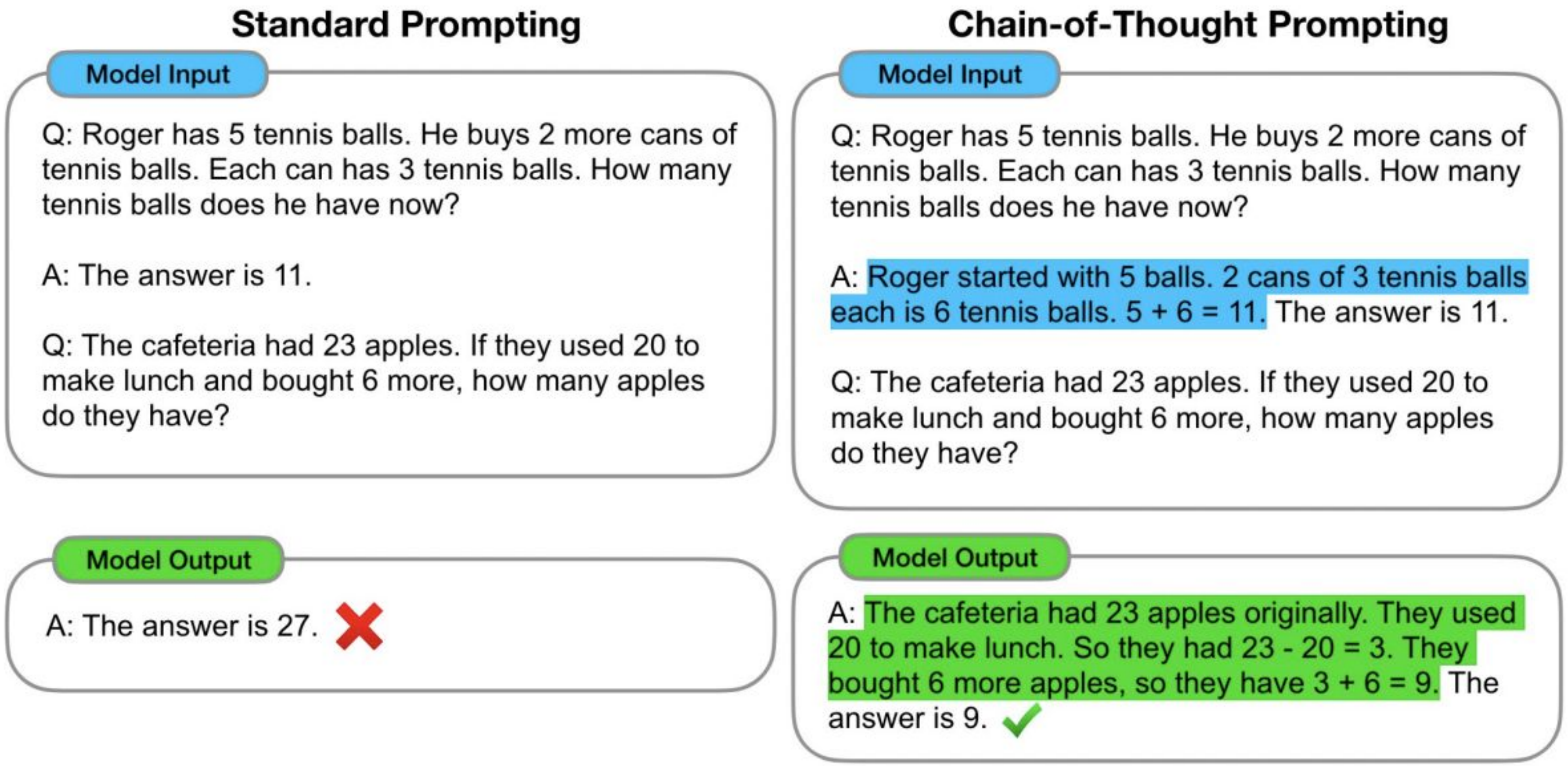
We hope you find these prompts useful and have fun using Claude!

Chain-of-Thought (CoT) Prompting



Chain-of-Thought (CoT) Prompting

- Asking the model to reason about its answer can improve its performance for few-shot in-context learning
- **Chain-of-thought prompting** provides such reasoning in the in-context examples



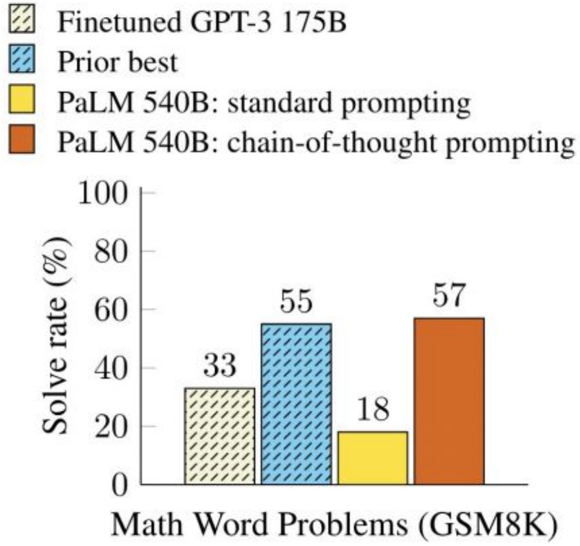


Figure 2: PaLM 540B uses chain-of-thought prompting to achieve new state-of-the-art performance on the GSM8K benchmark of math word problems. Finetuned GPT-3 and prior best are from Cobbe et al. (2021).

Chain-of-Thought (CoT) Prompting

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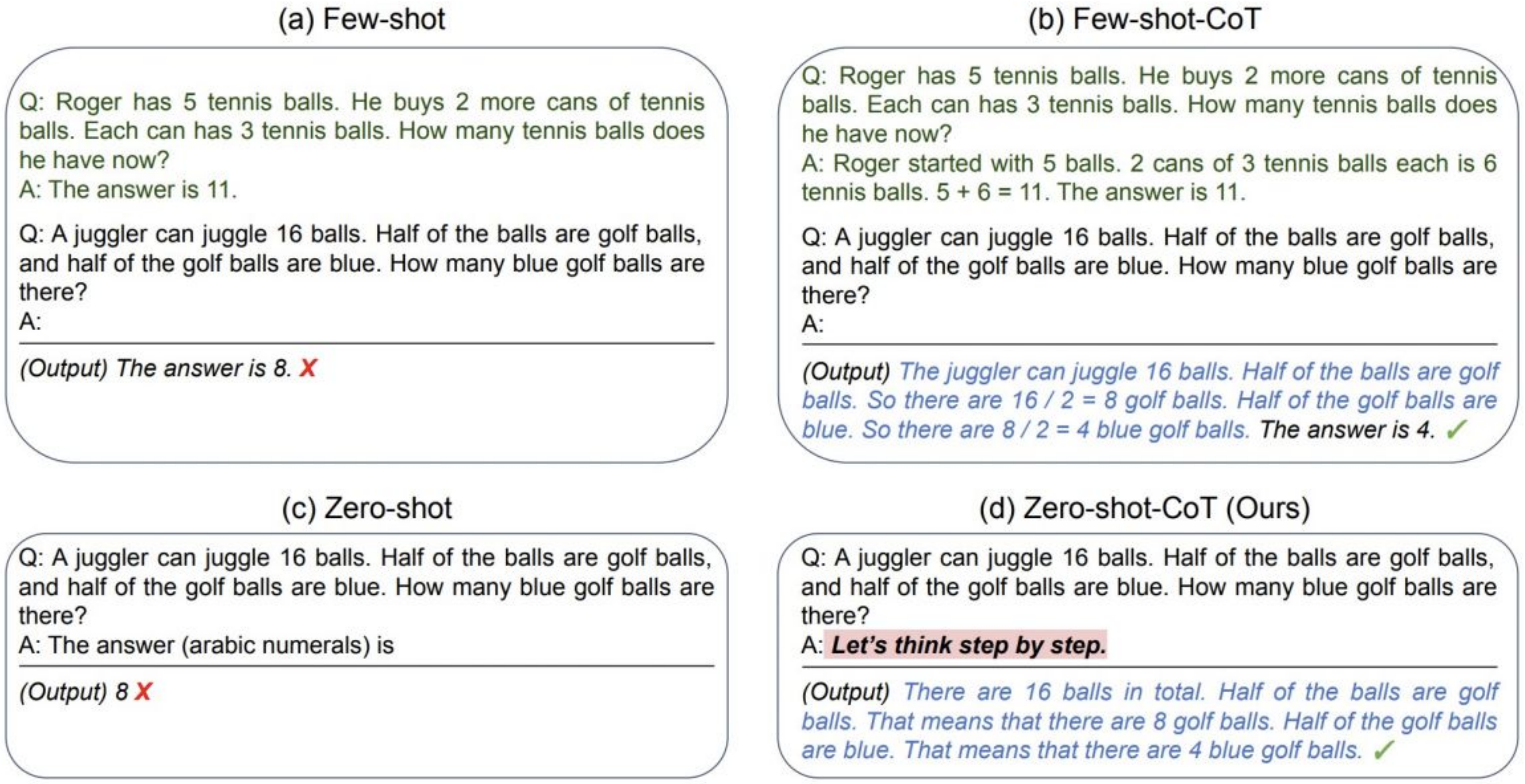

- But the model does better even if you just prompt it to reason step-by-step

Chain-of-Thought (CoT) Prompting

- Asking the model to reason about its answer can improve its performance for few-shot in-context learning
- **Chain-of-thought prompting** provides such reasoning in the in-context examples

	MultiArith	GSM8K
Zero-Shot	**17.7**	**10.4**
Few-Shot (2 samples)	33.7	15.6
Few-Shot (8 samples)	33.8	15.6
Zero-Shot-CoT	**78.7**	**40.7**
Few-Shot-CoT (2 samples)	84.8	41.3

- But the model does better even if you just prompt it to reason step-by-step

Thanks

Q&A

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